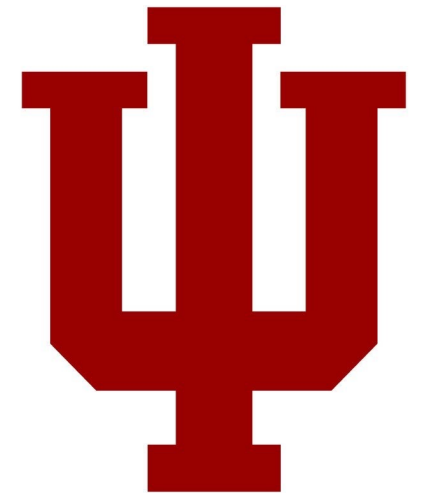


# 01: Standard Cells

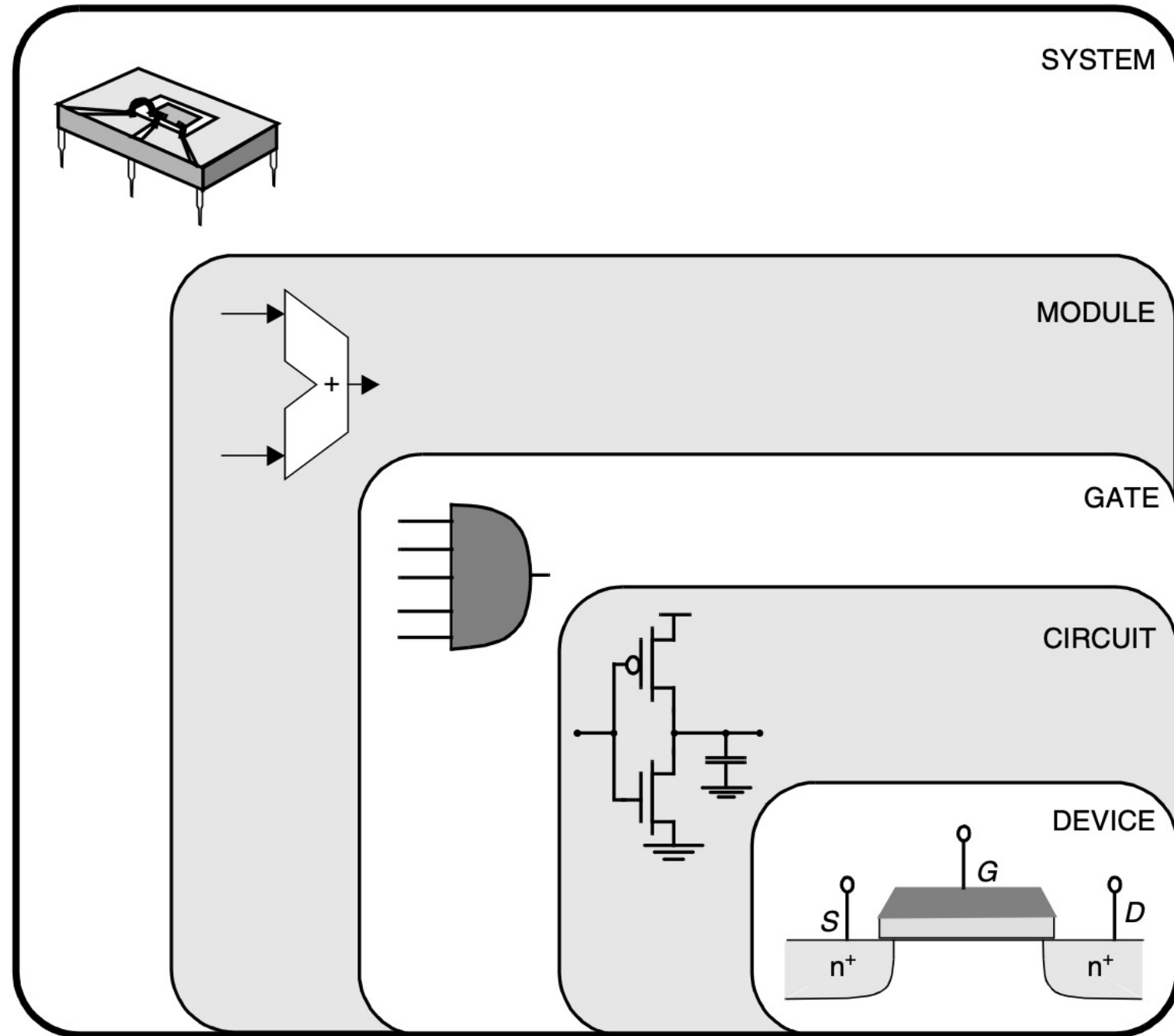
E599– Digital Integrated Circuit Design



Website

<https://engr599-ic.github.io>

# Review Time

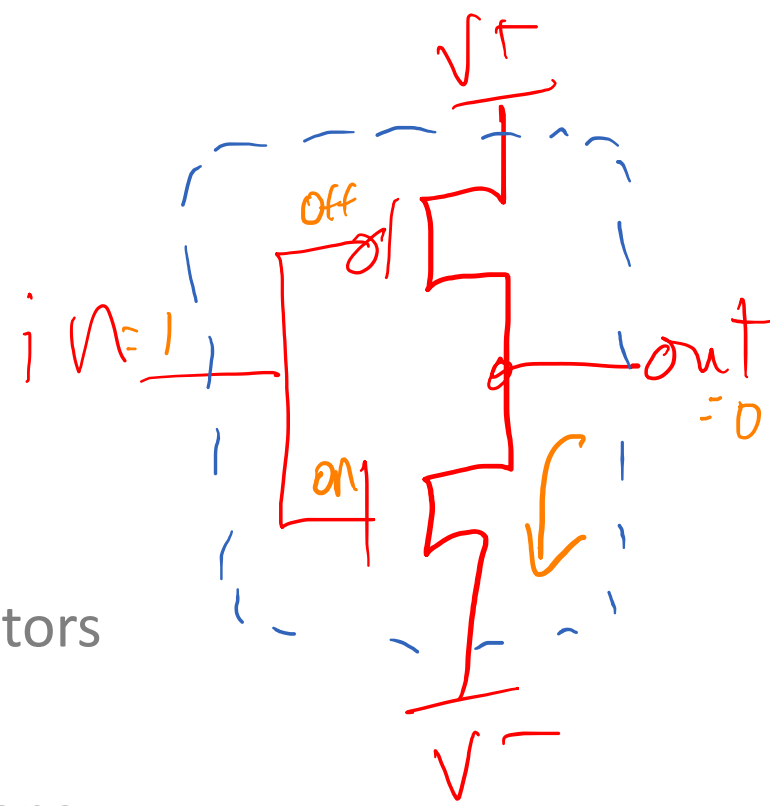


# Devices Review

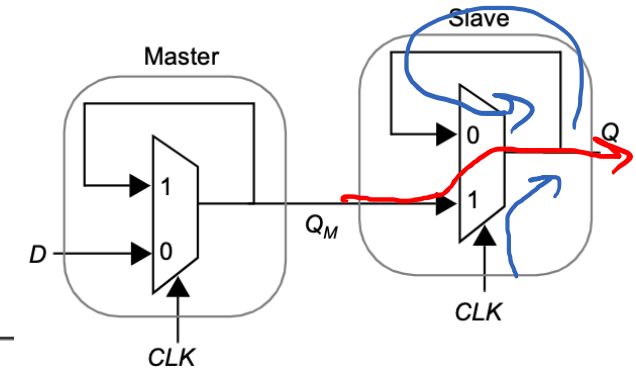
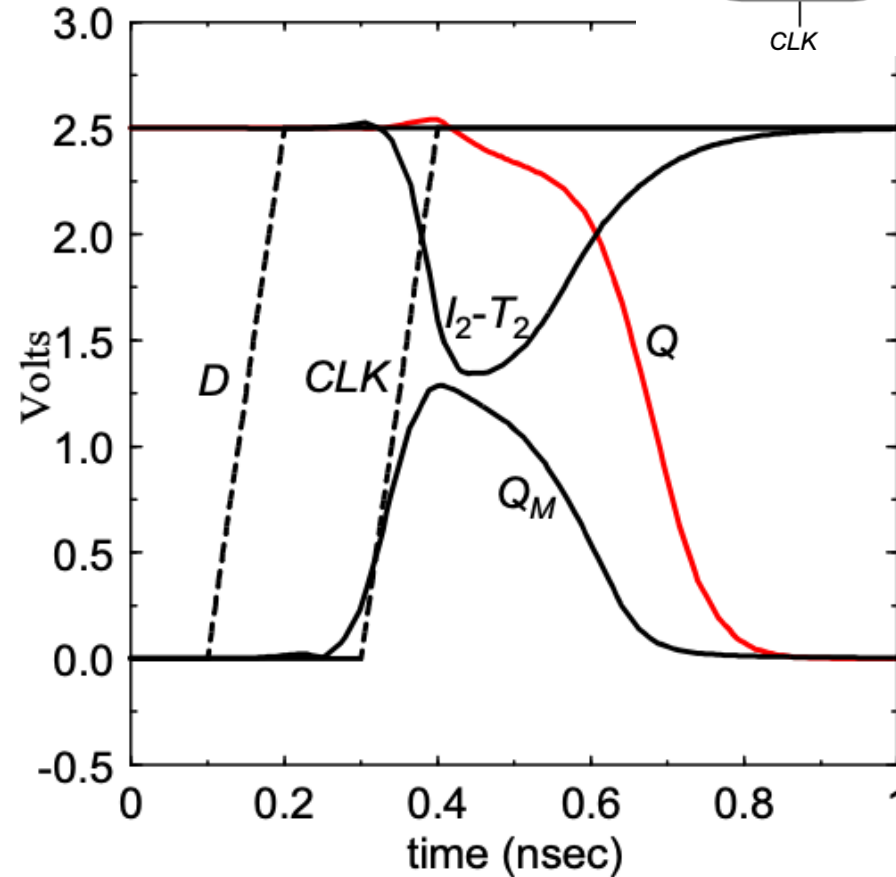
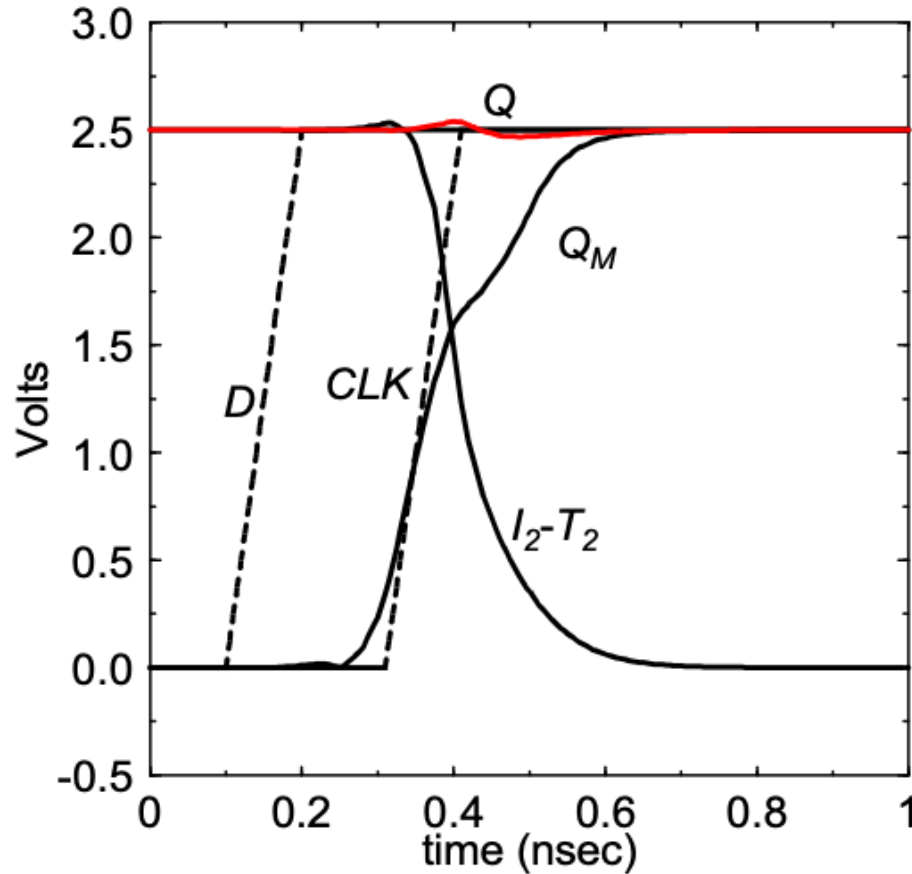
- Transistors
  - Voltage on gate causes conduction between source + drain. Aka “Magic”
  - Only consider fully-off or fully-on (saturated)
  - Two types: NMOS and PMOS
    - PMOS: 0V=on
    - NMOS: 0V =off
- Wires
  - Resistor + Capacitor
  - Longer wires = slower signals

# Combinational Circuits Review

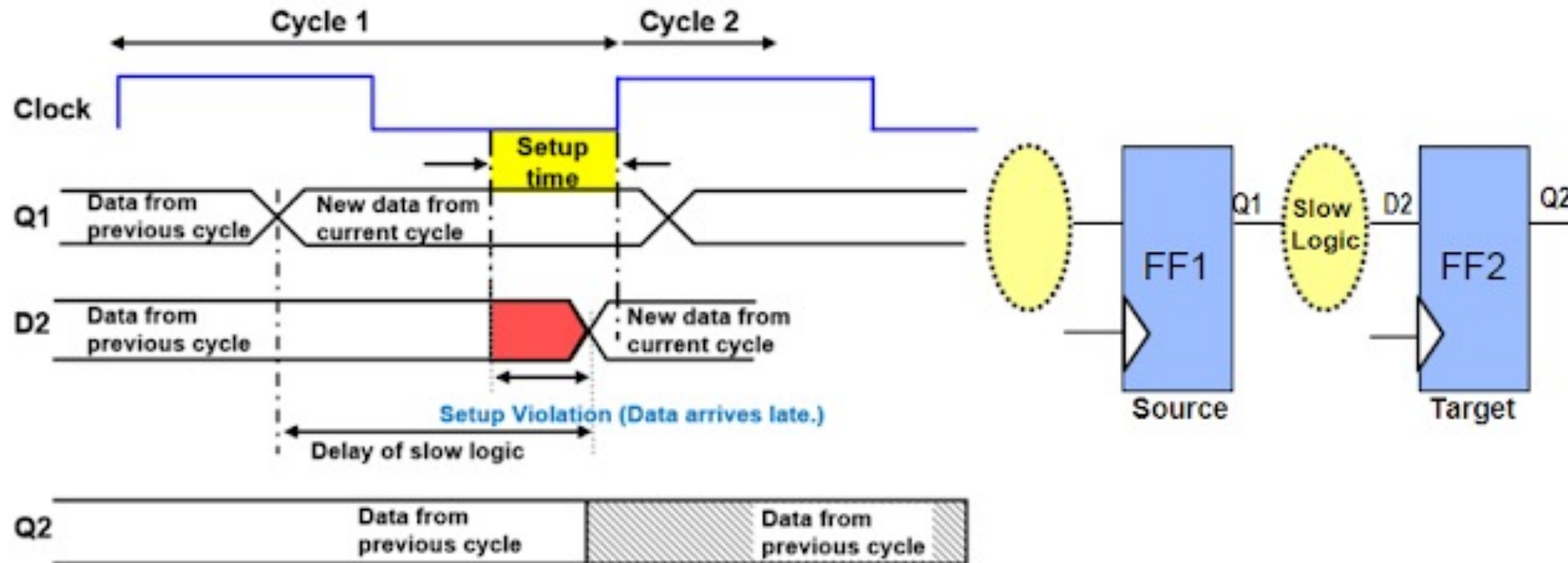
- Inverter: 1 NMOS + 1 PMOS
- Process Variation:
  - Some faster, some slower transistors
- Intrinsic Delay:
  - Fundamental delay in all transitions
  - Bigger transistors reduce delays, to a point.
- Other Delays
  - output takes times to reflect input changes
  - Bigger capacitive loads increase delays



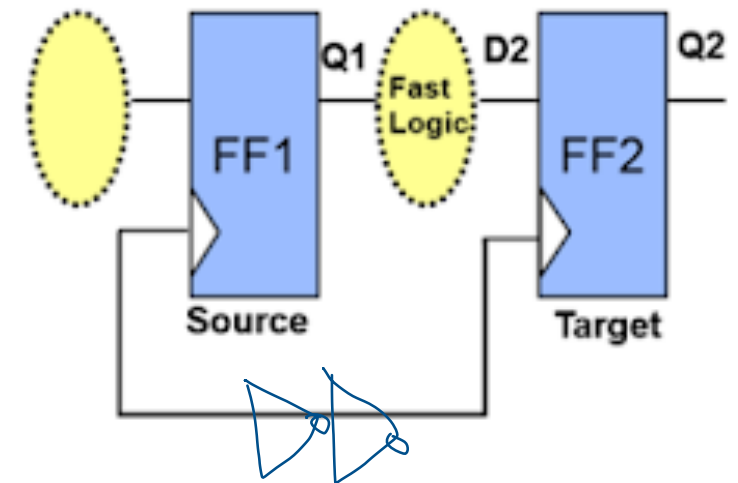
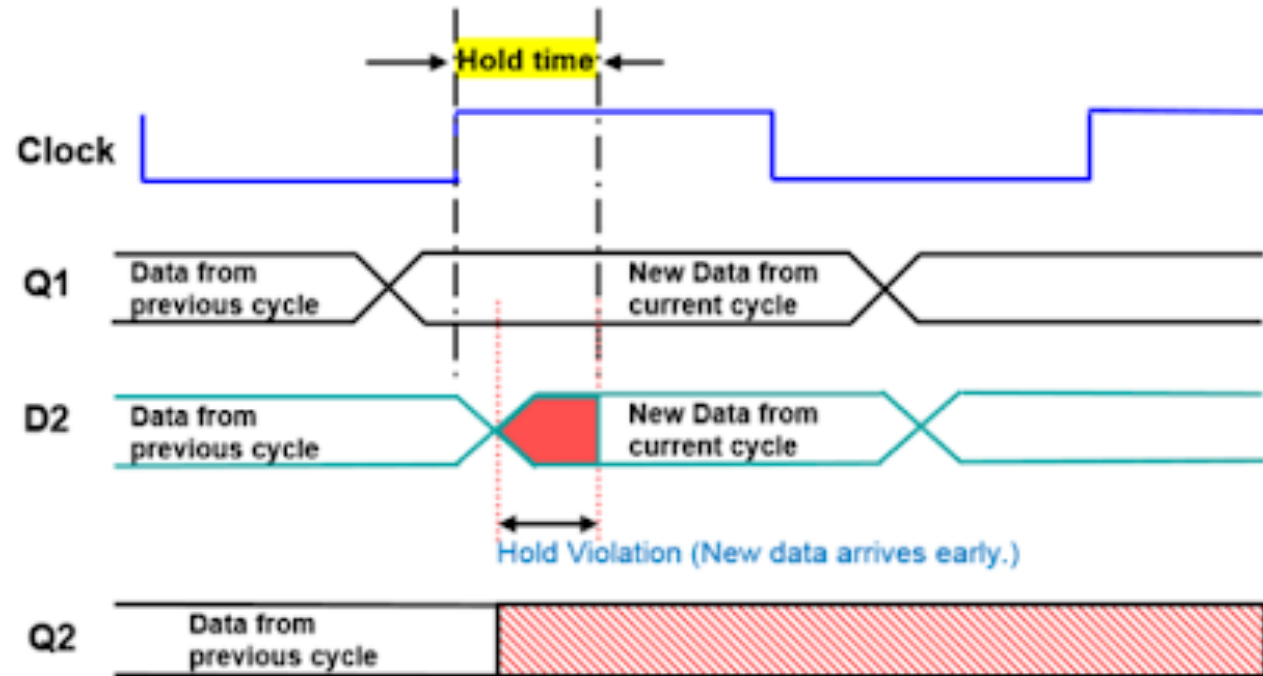
# Setup Time Violations



# Setup Time

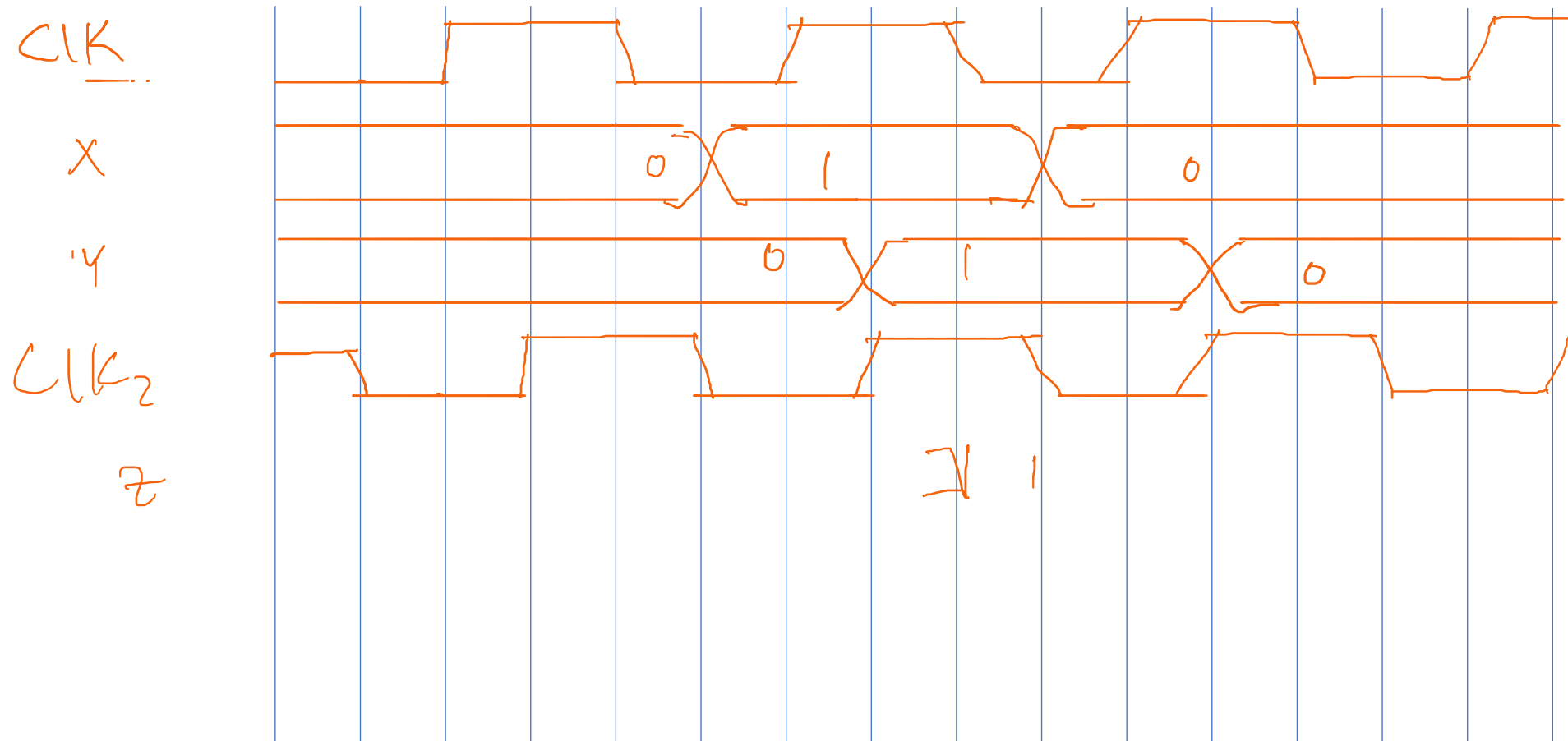
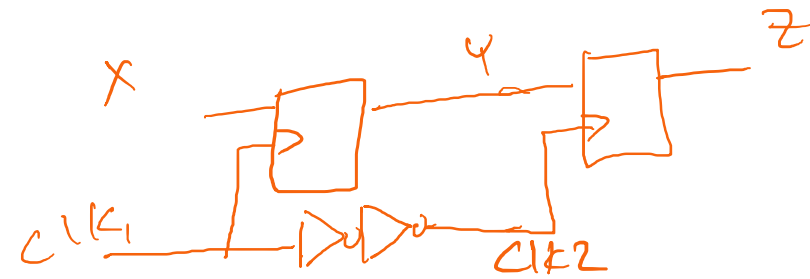


# Hold Time

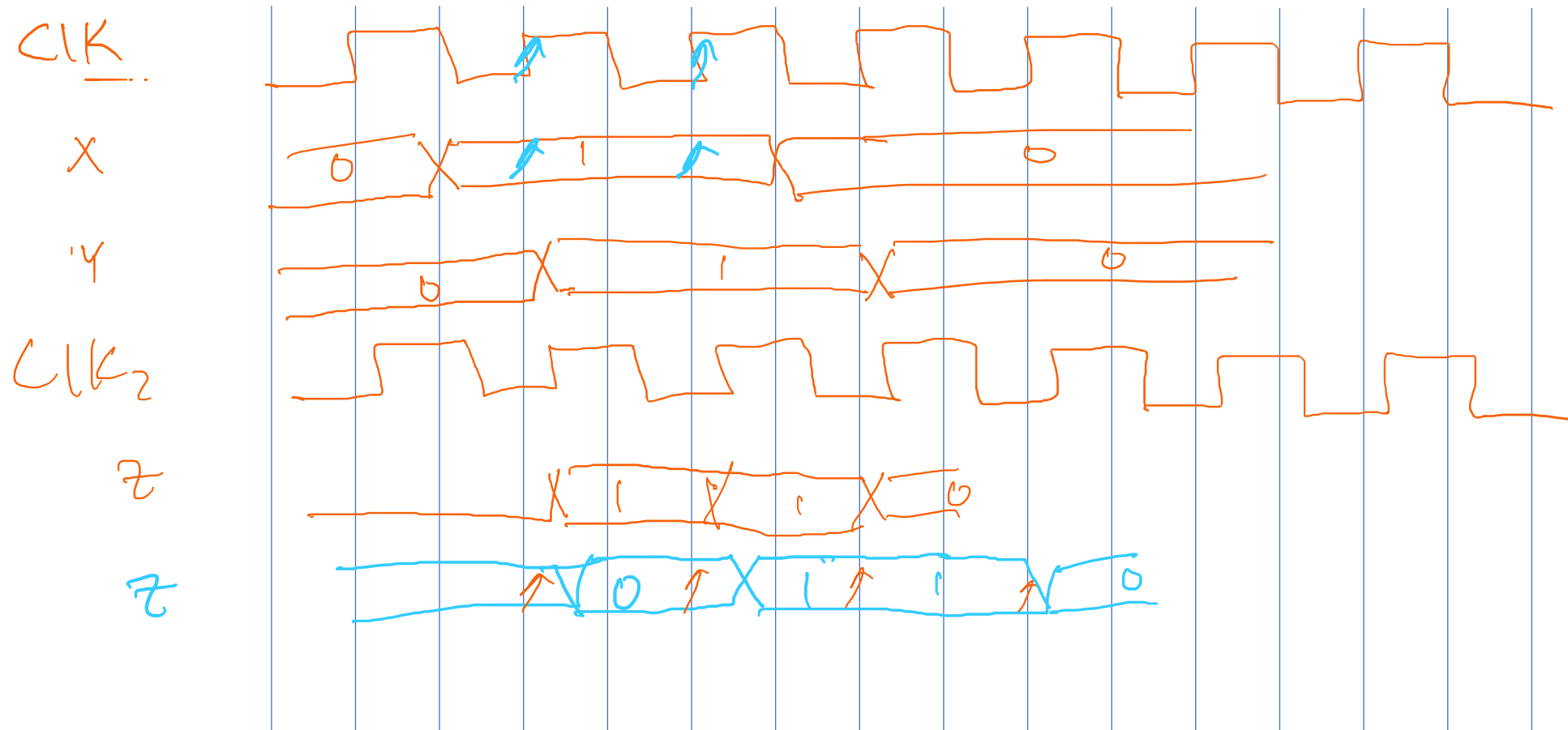
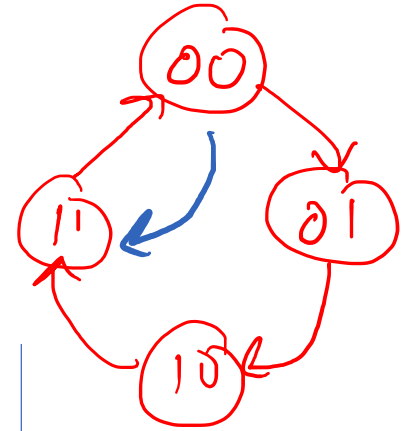
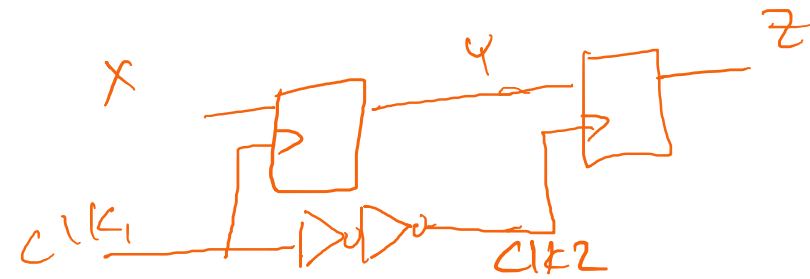




# Hold Time Diagram



# Hold Time Diagram



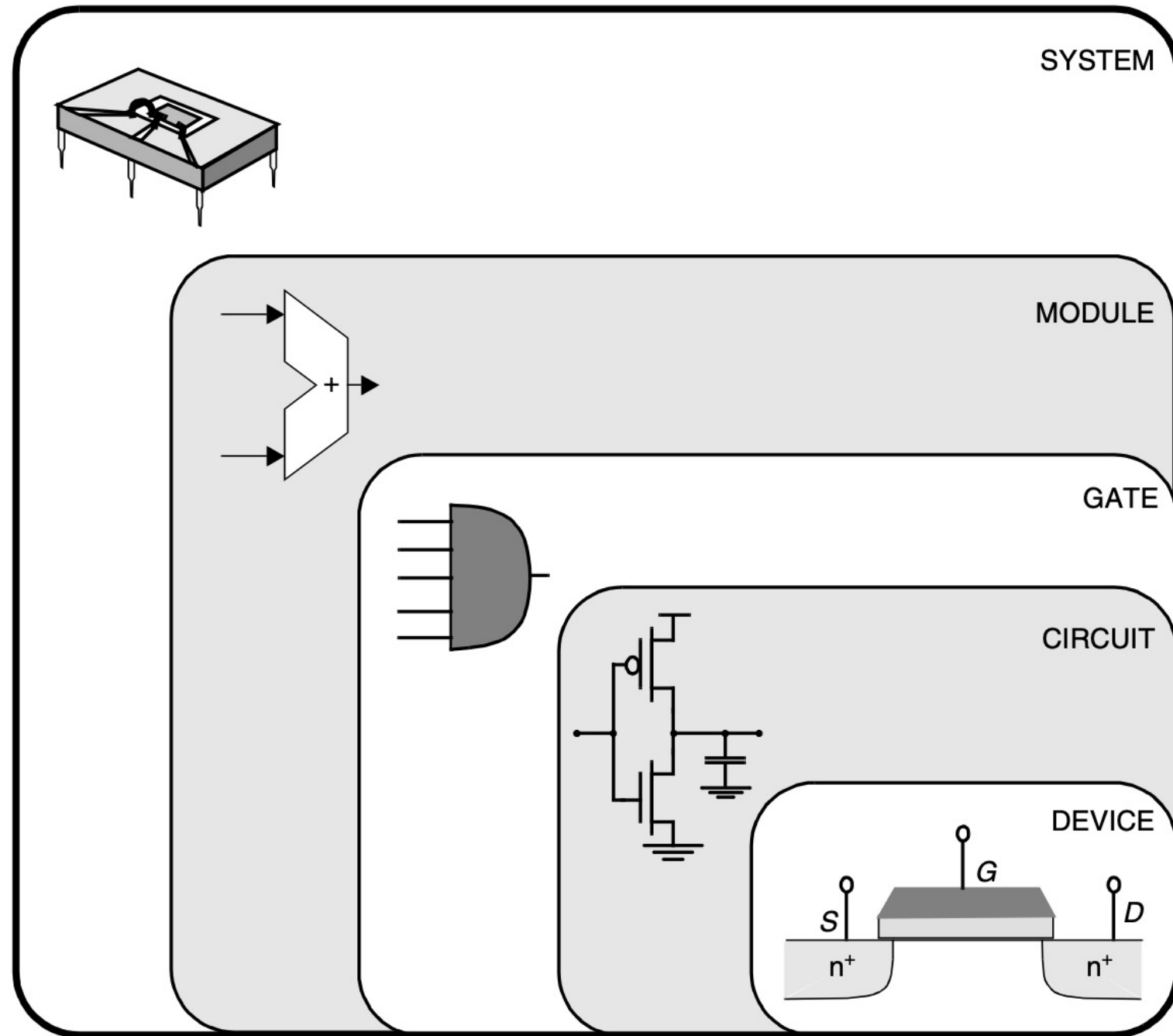
# WARNING: HOLD TIMES ARE A BIG DEAL!!!

- You can always fix setup times with a slower clock.
- Hold times are unfixable....
- You do not want a hold-time violation in your design!

# Sequential Circuits Review

- Latches: Output follows input when clock is high
- Flip-Flop: Output follows input on the rising edge of the clock
- Setup Time: Time input must be stable before the clock rising edge
- Hold Time: Time input must be stable after the clock rising edge
- Setup affects performance. **Hold affects correctness.**

# This Time: Gates

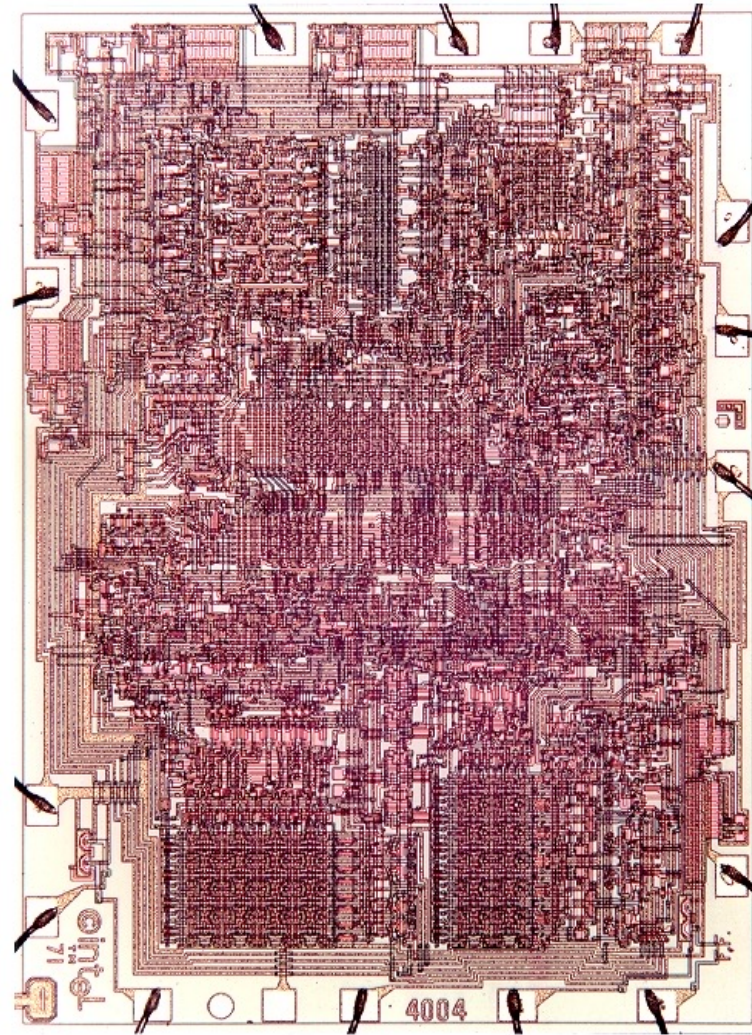


# “Standard” Cell Design

Std Cell

Std Lib

# The “Full-Custom” Approach

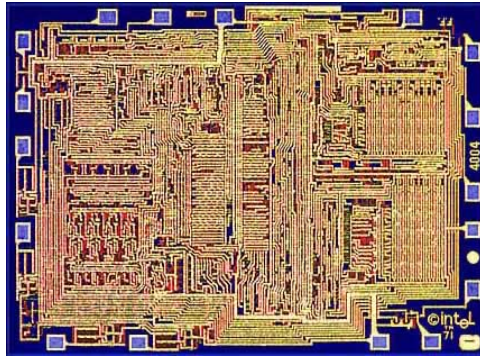


Intel 4004

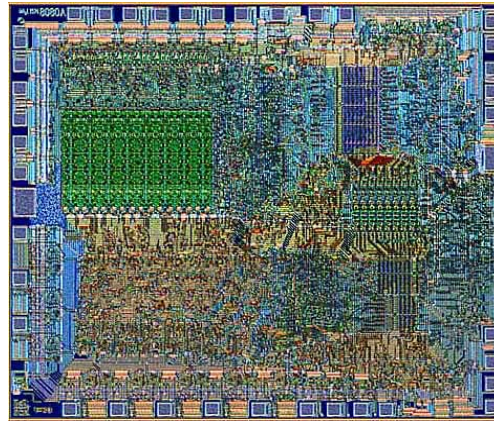
Courtesy Intel



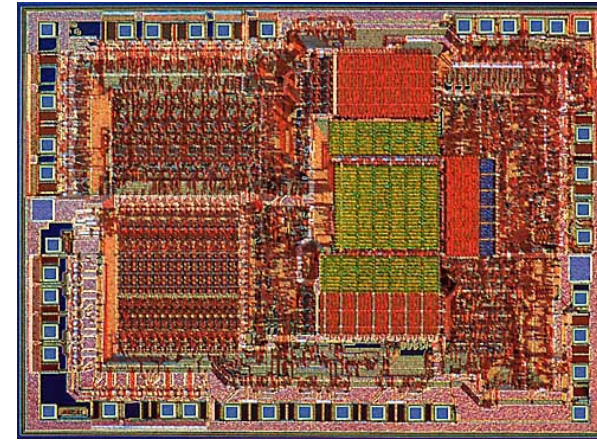
# Transition to Automation and Regular Structures



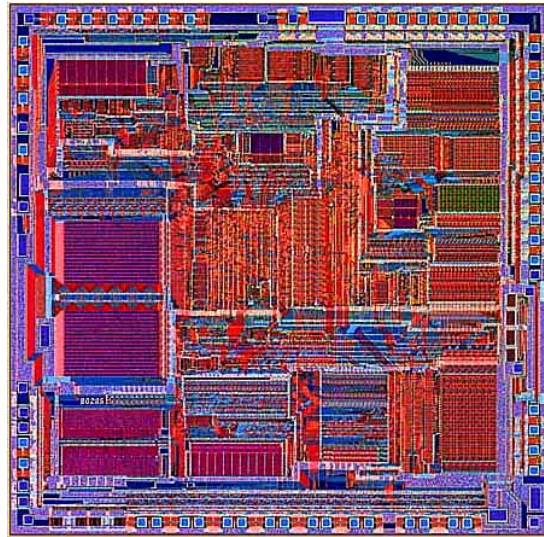
Intel 4004 ('71)



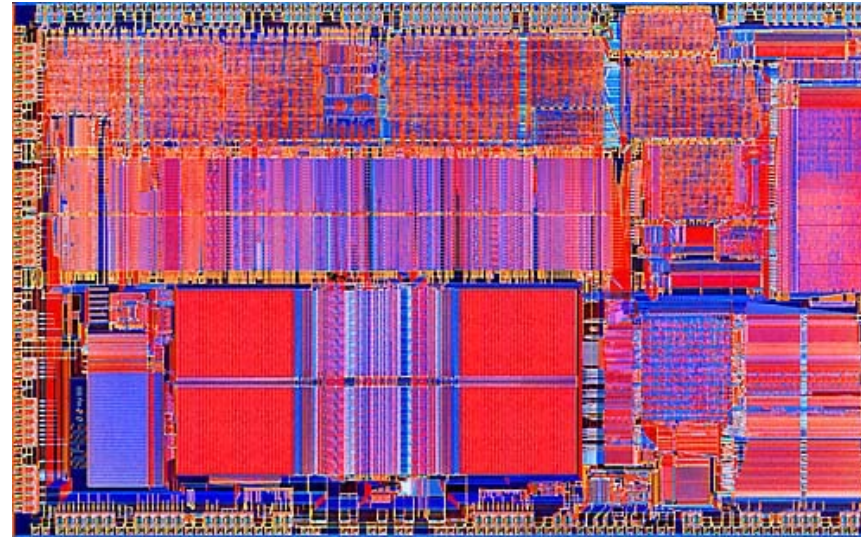
Intel 8080



Intel 8085



Intel 8286

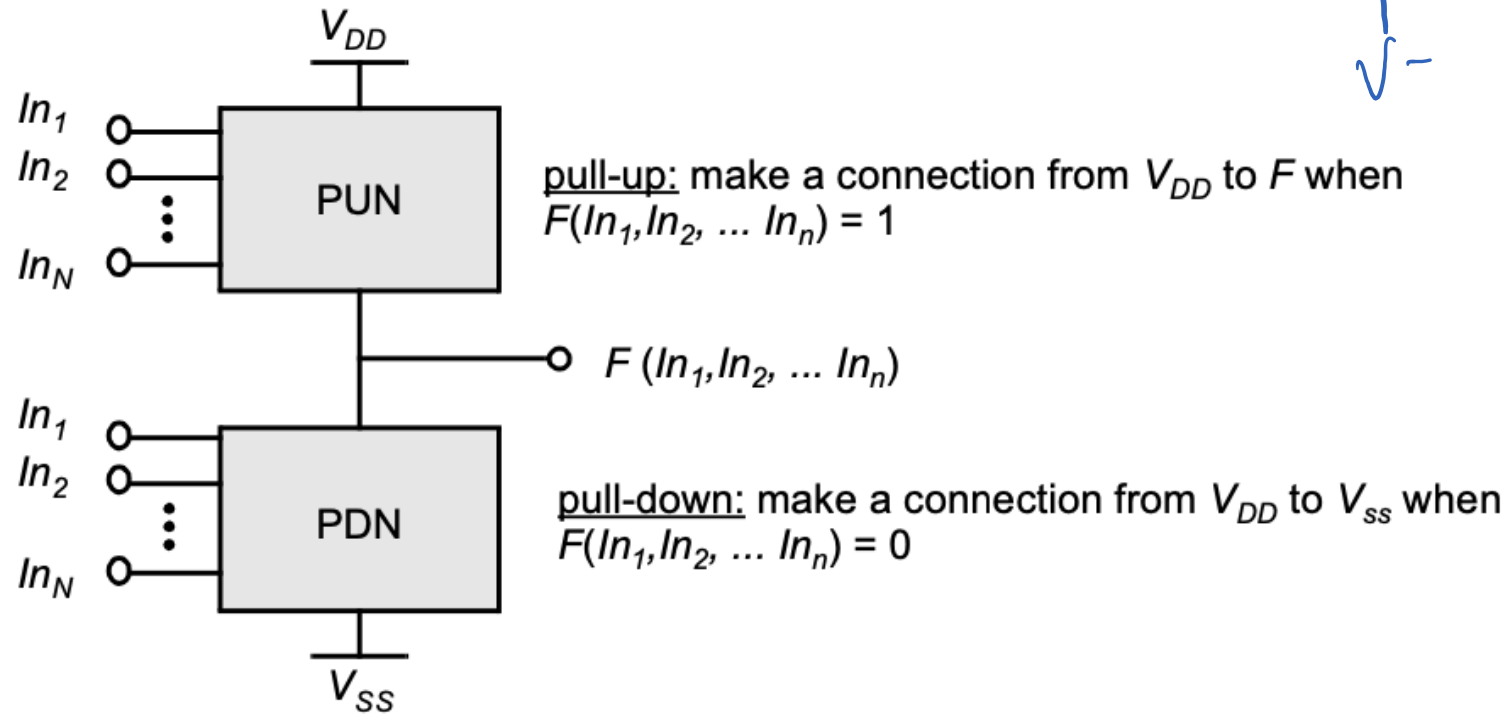
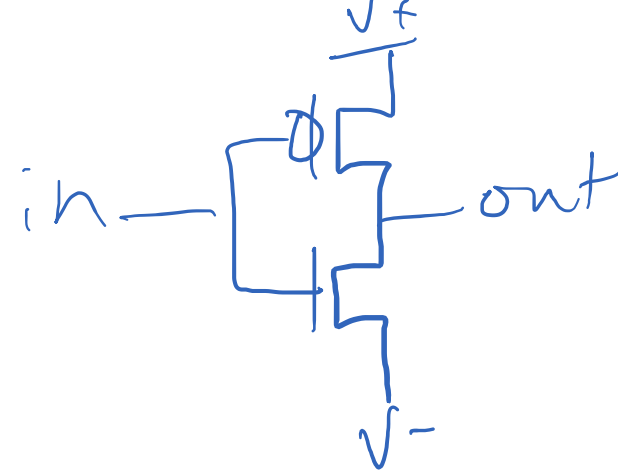


Intel 8486

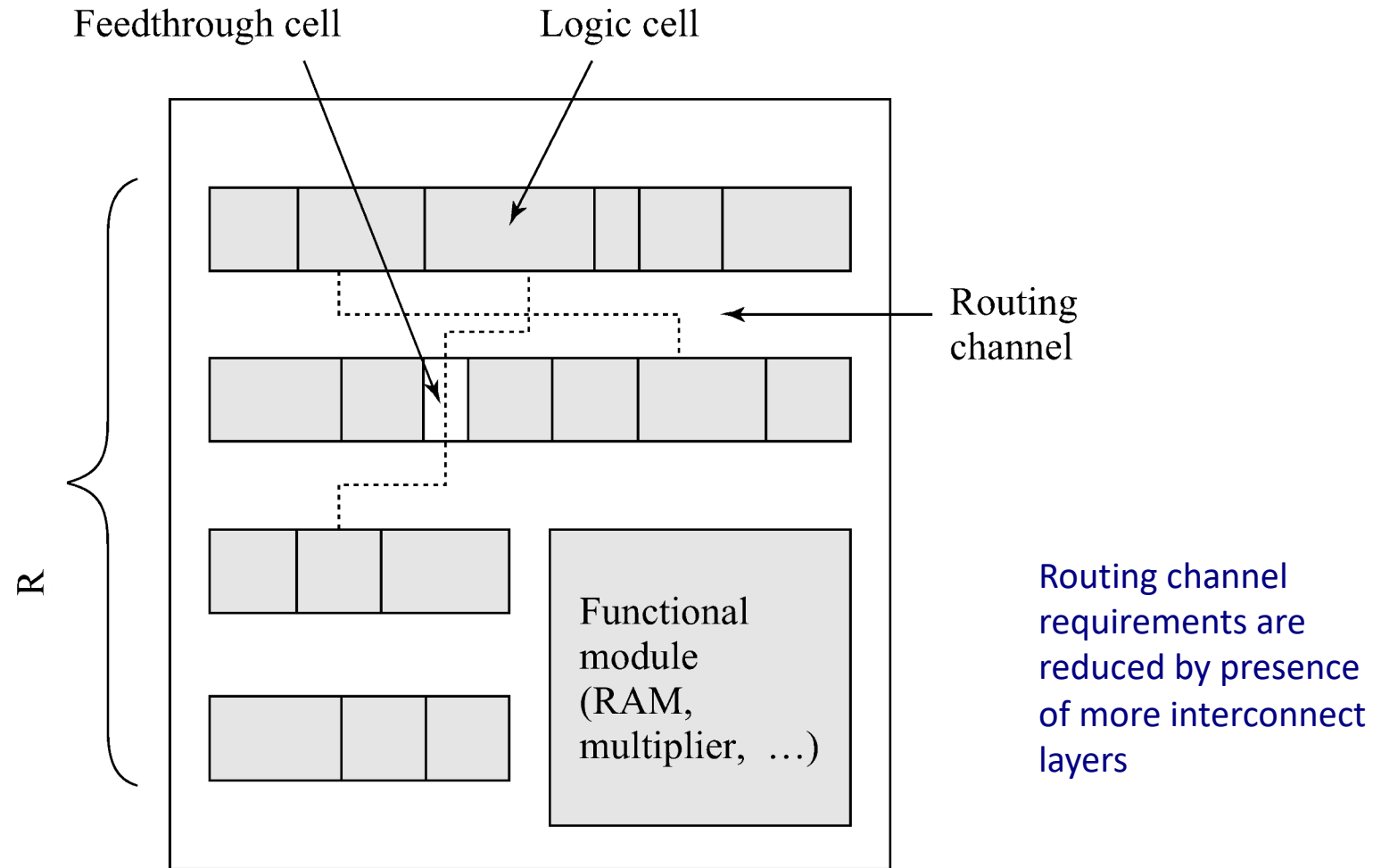
Courtesy Intel



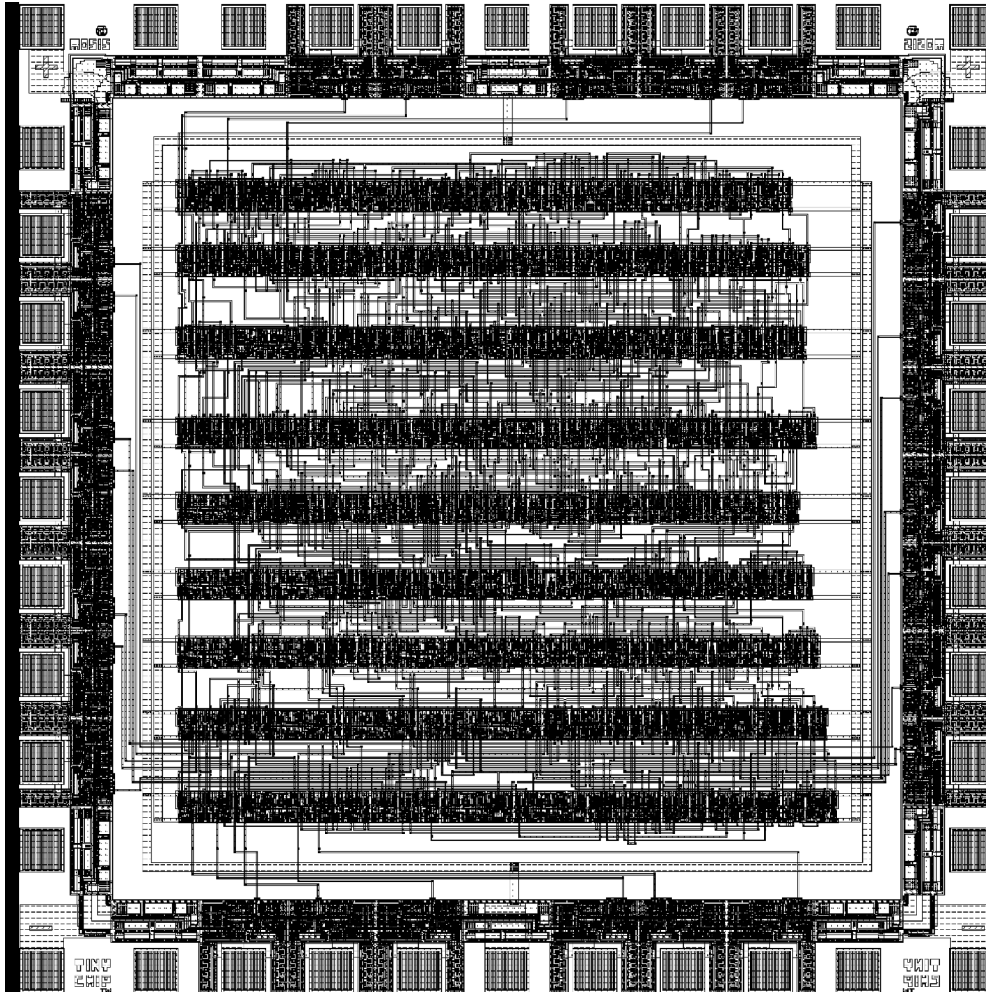
# Standardized Logic Gates



# Cell-based Design (or standard cells)

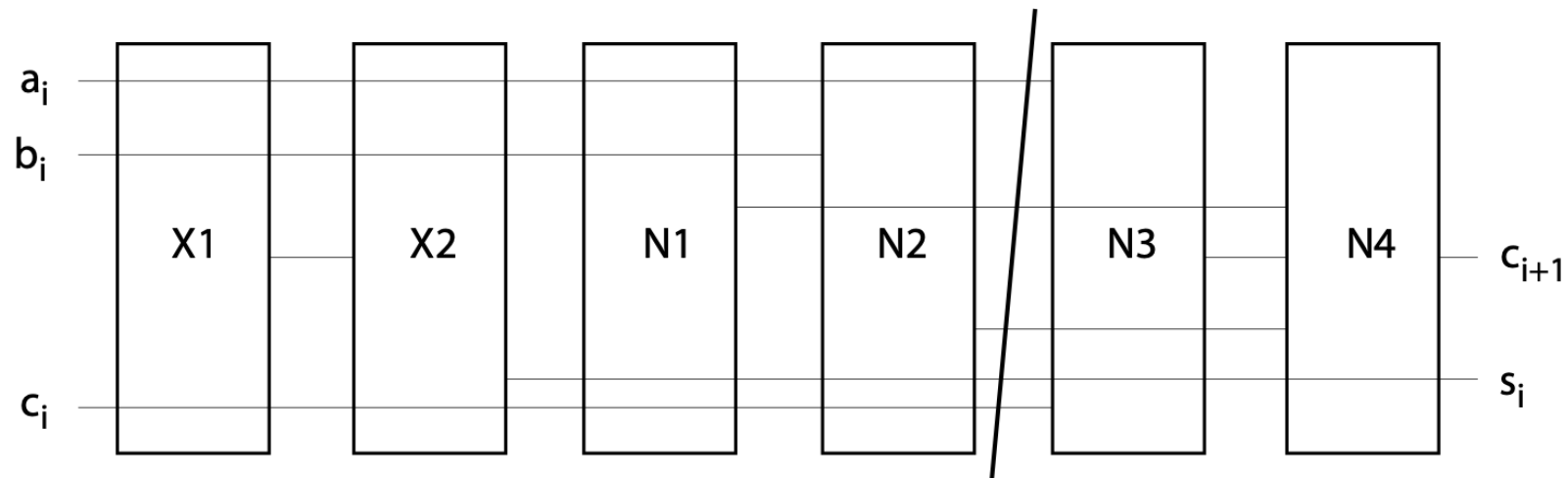


# Standard Cell — Example

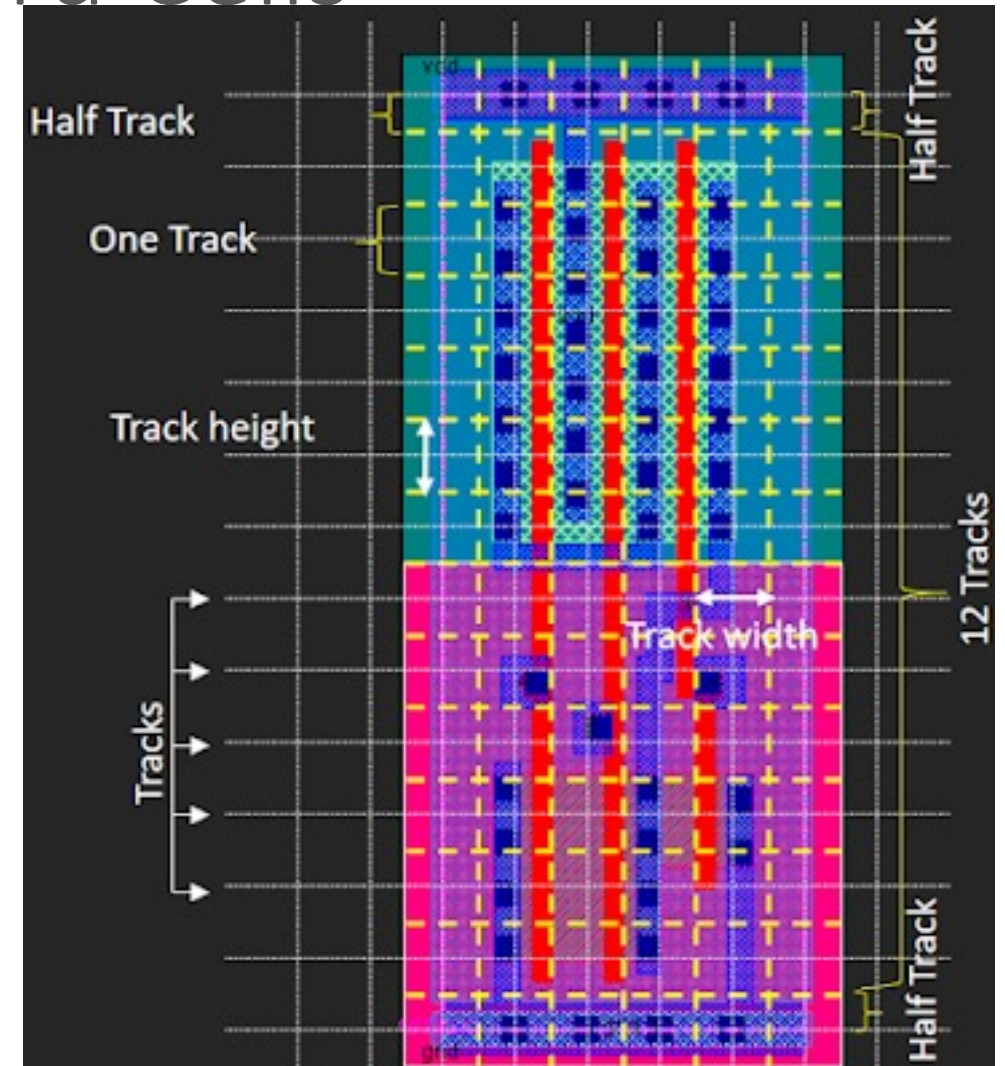
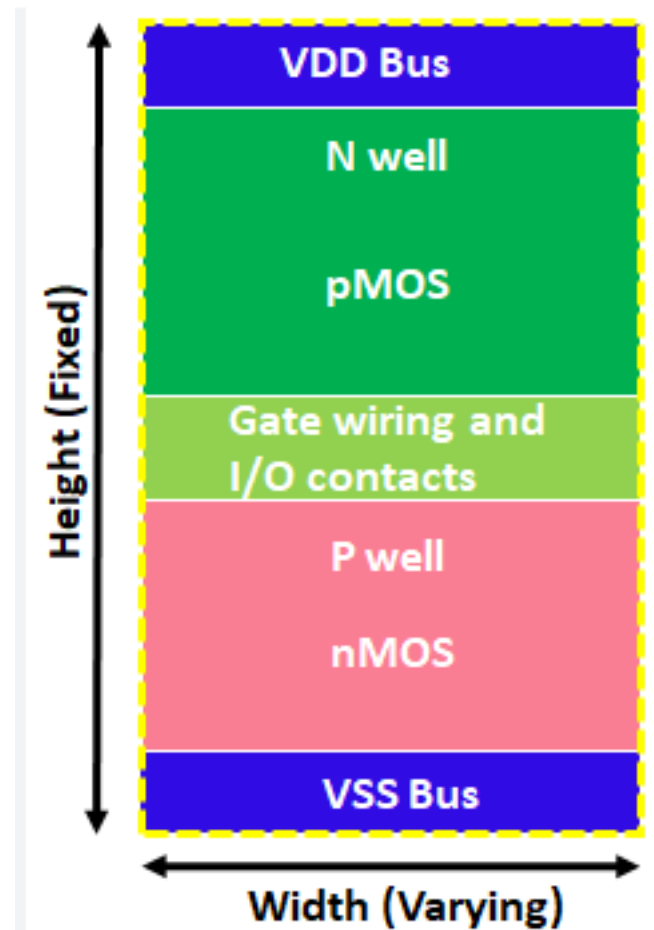


[Brodersen92]

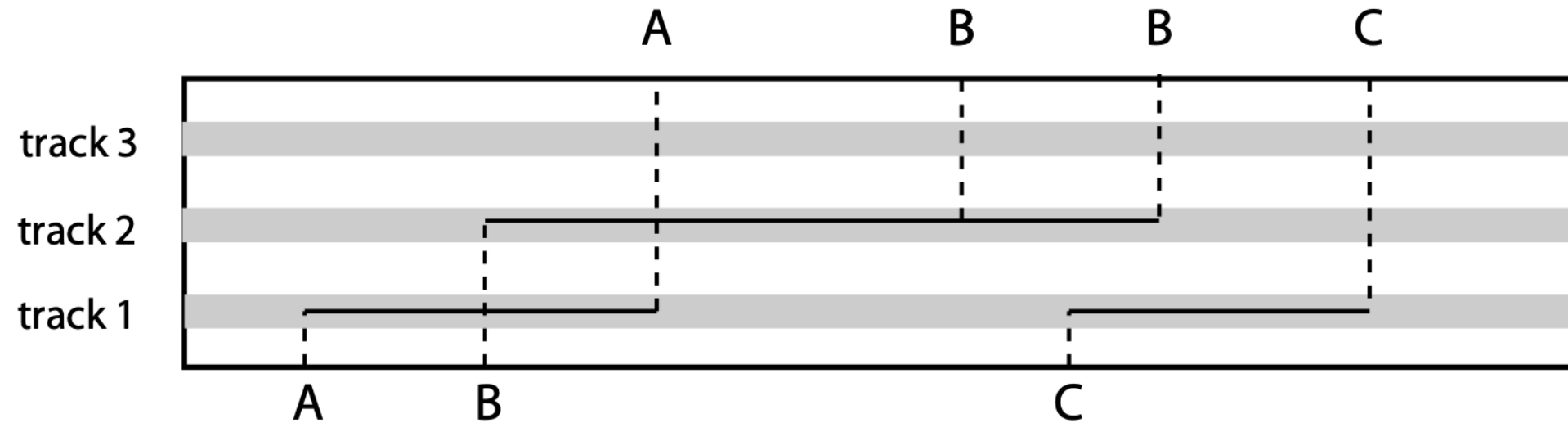
Put the wires above the logic cells



# Routing Tracks w/in Standard Cells

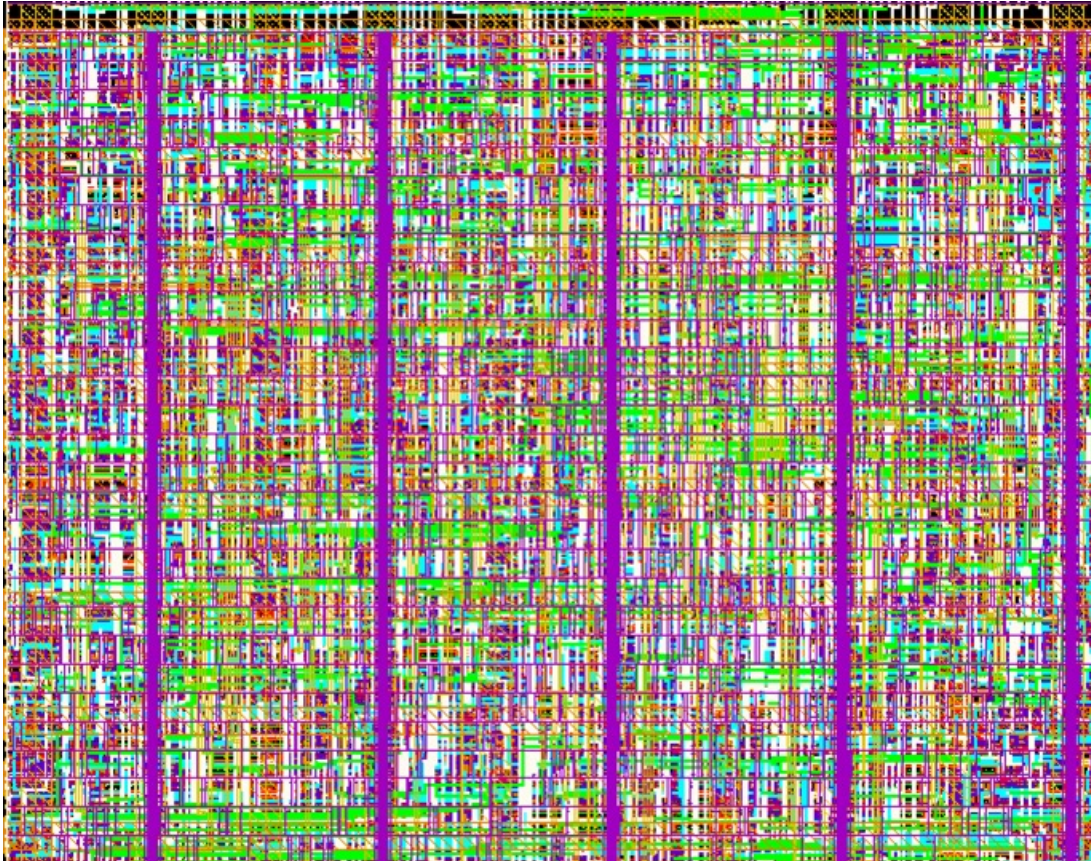


# Routing Tracks



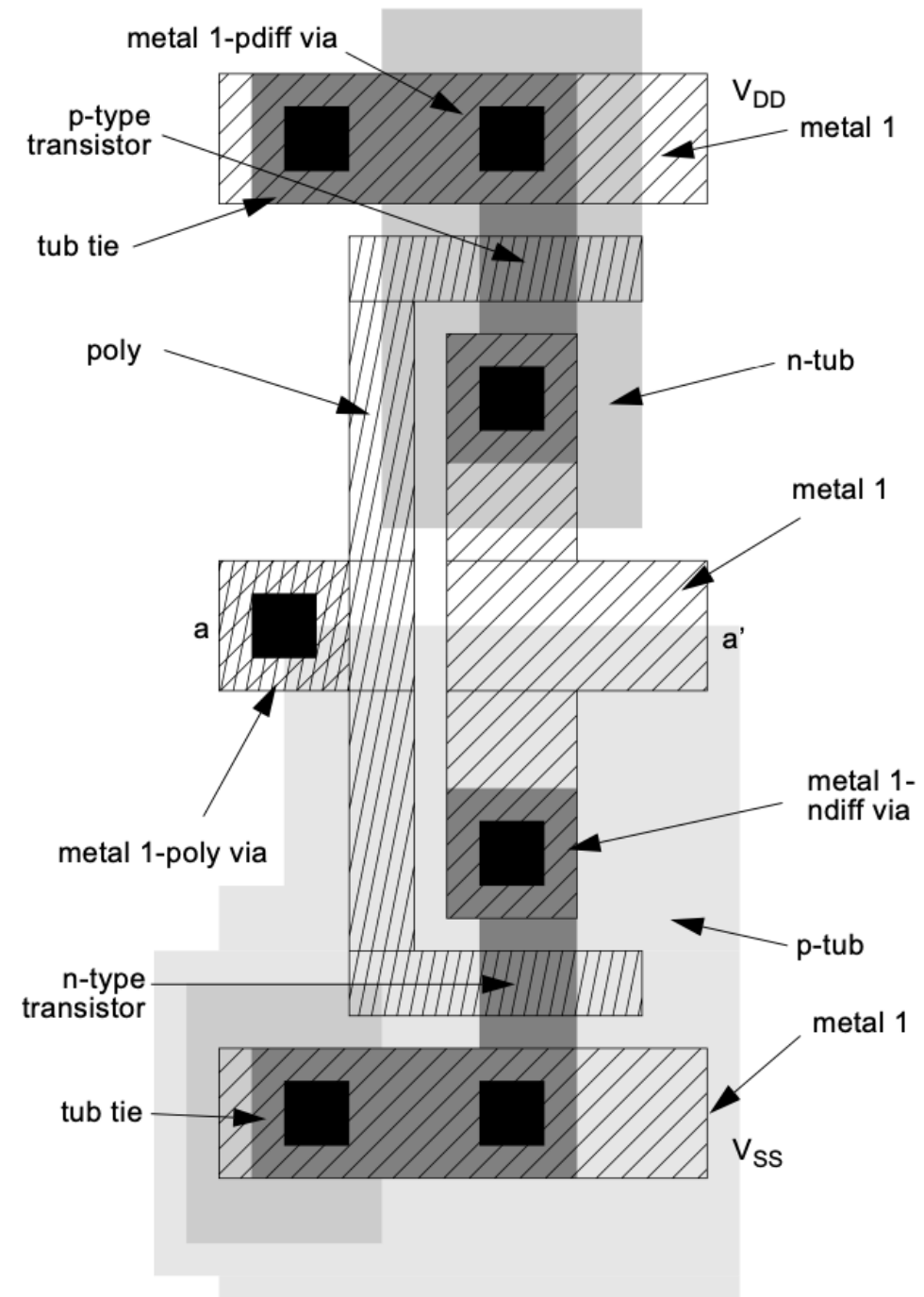


# Standard Cell – The New Generation

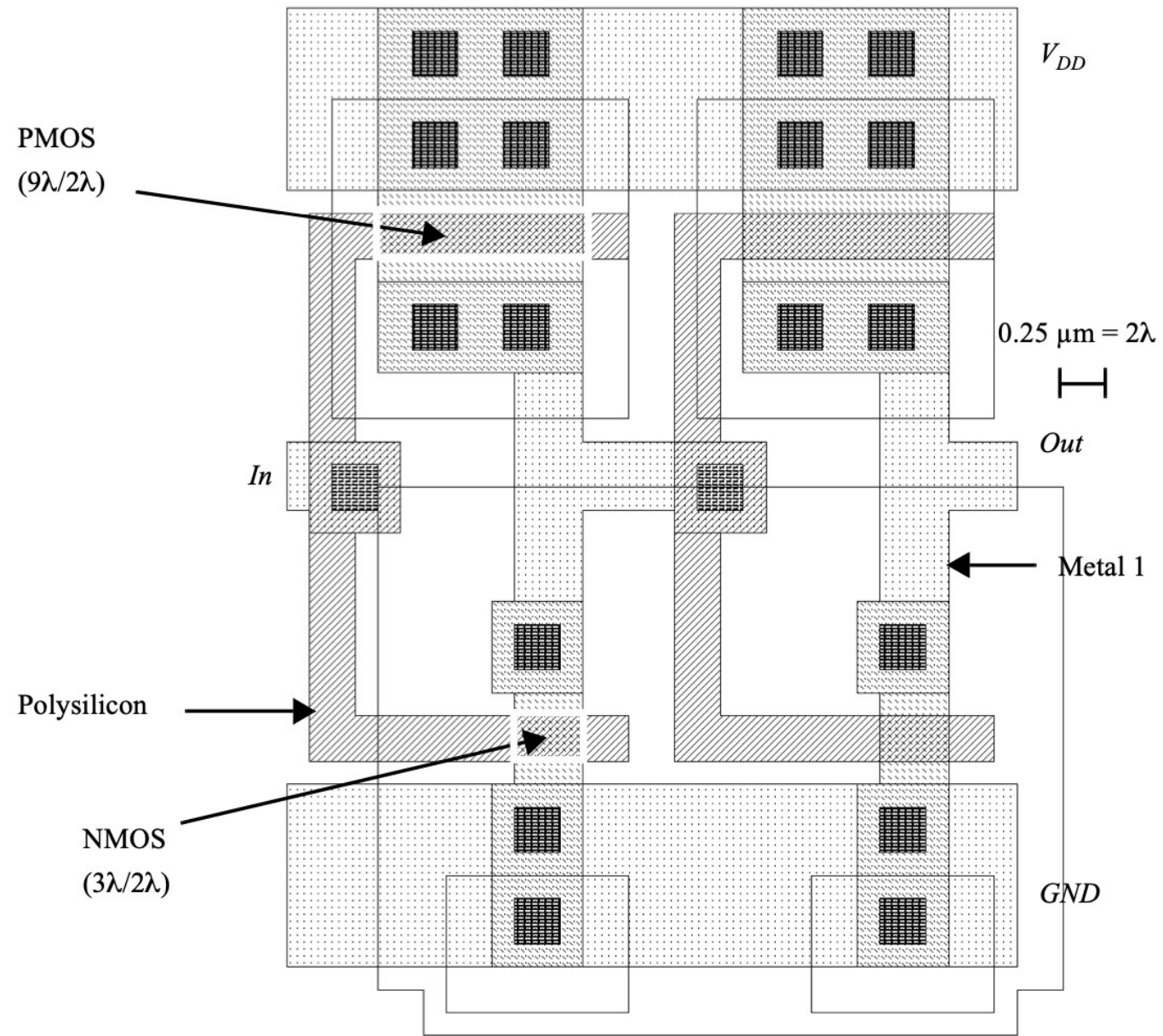


Cell-structure  
hidden under  
interconnect layers

# Inverter Layout





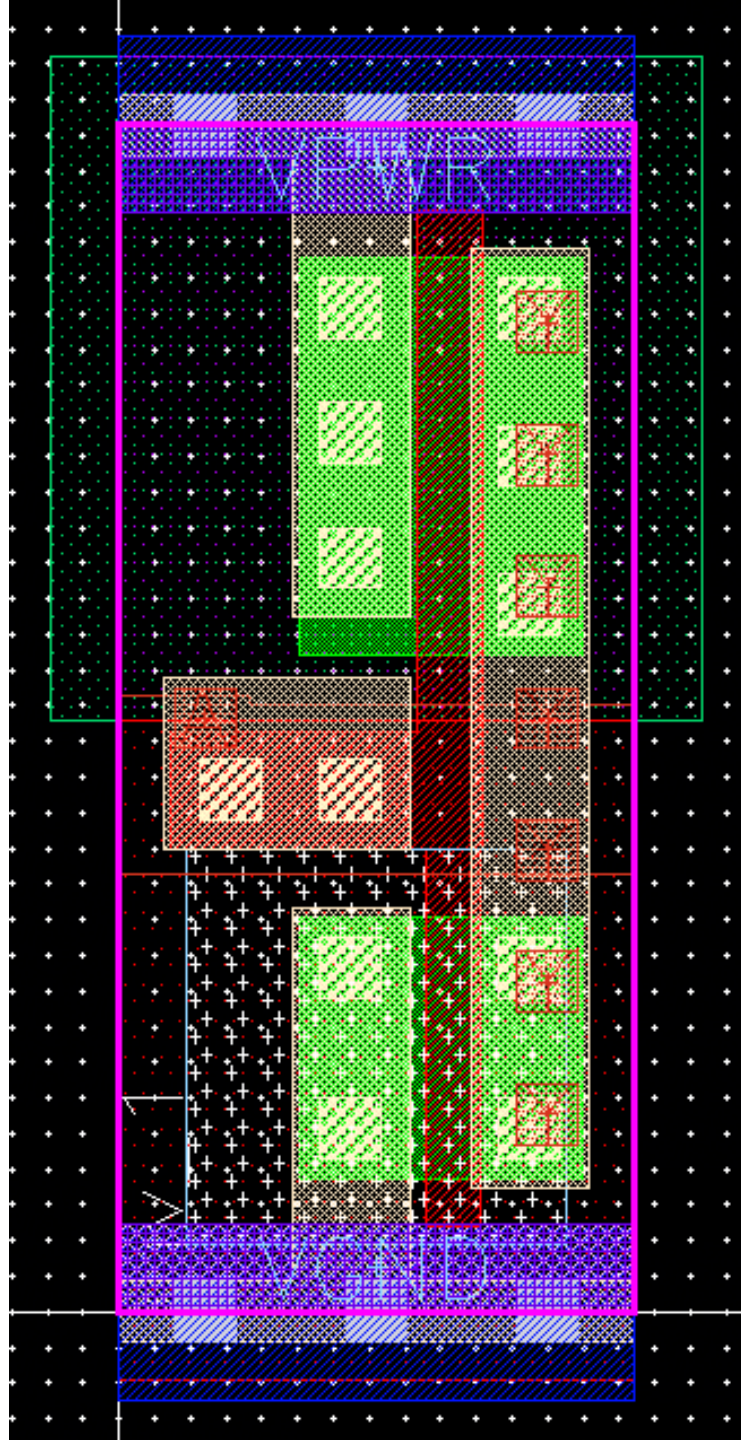


[Digital Integrated  
Circuits: A Design  
Perspective]

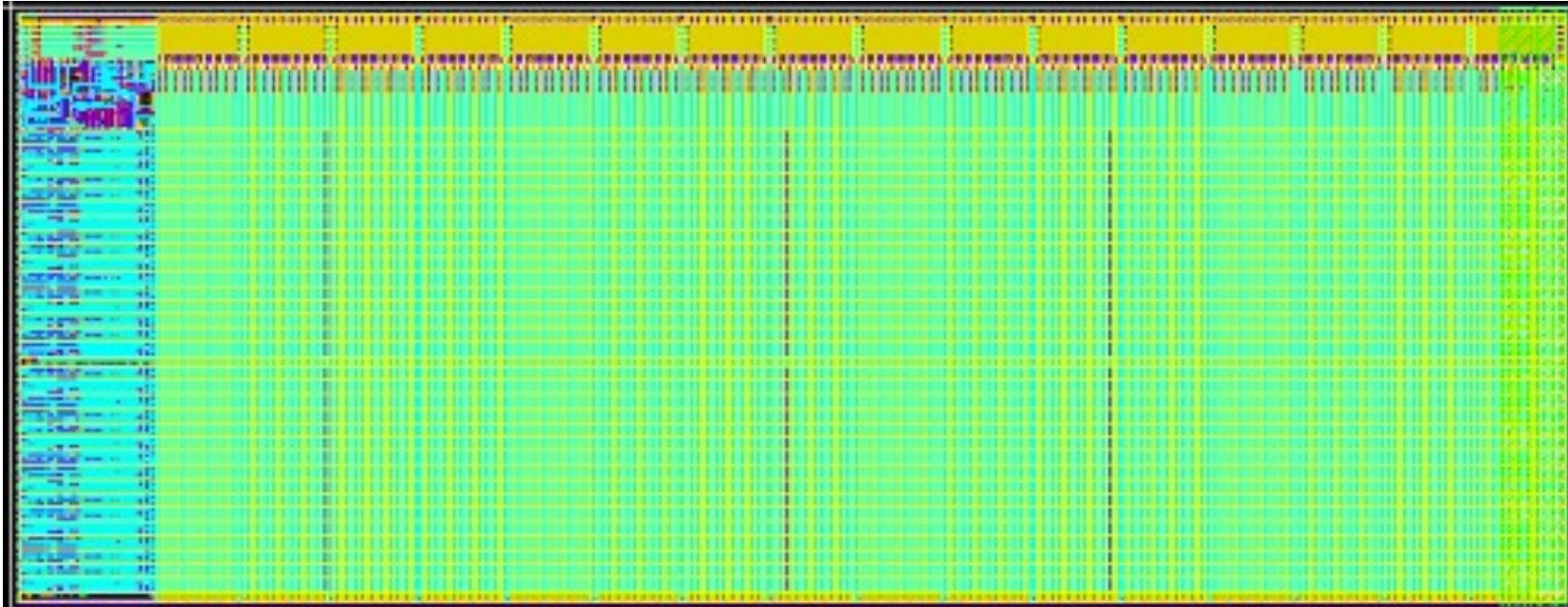
**Figure 5.15** Layout of two chained, minimum-size inverters using SCMOS Design Rules (see also Color-plate 6).

# Our INV\_X1

[demo time]



# “Hard” MacroModules

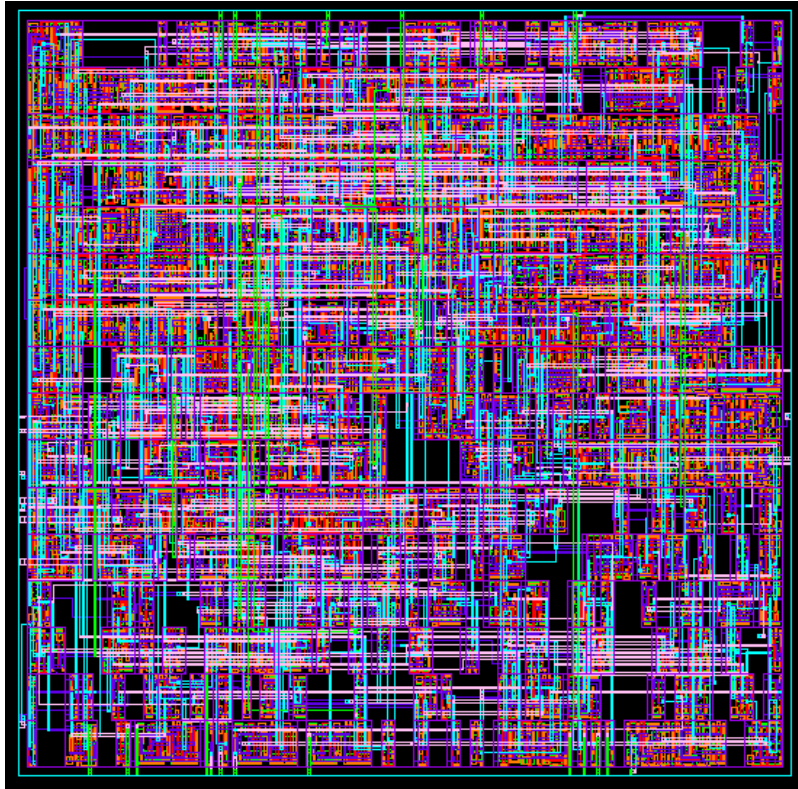


256×32 (or 8192 bit) SRAM

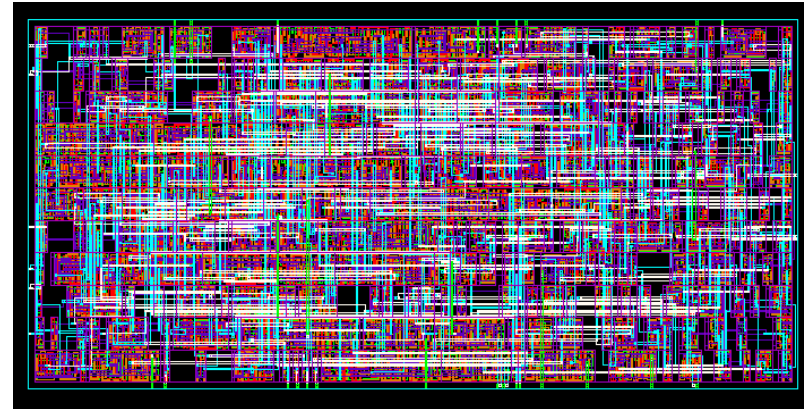
Generated by hard-macro module generator



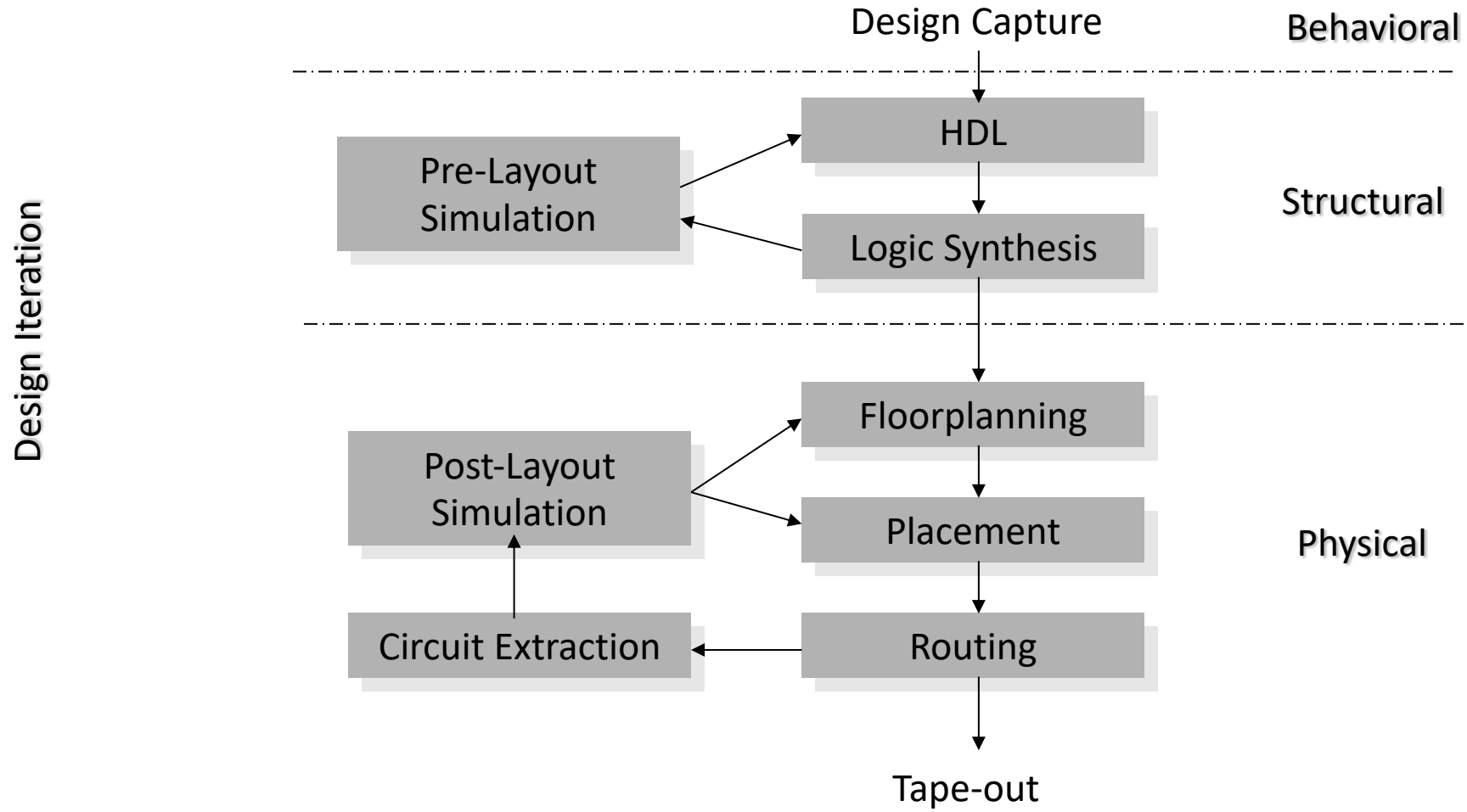
# “Soft” MacroModules



```
string mat = "booth";  
directive (multtype = mat);  
output signed [16] Z = A * B;
```



# Semicustom Design Flow

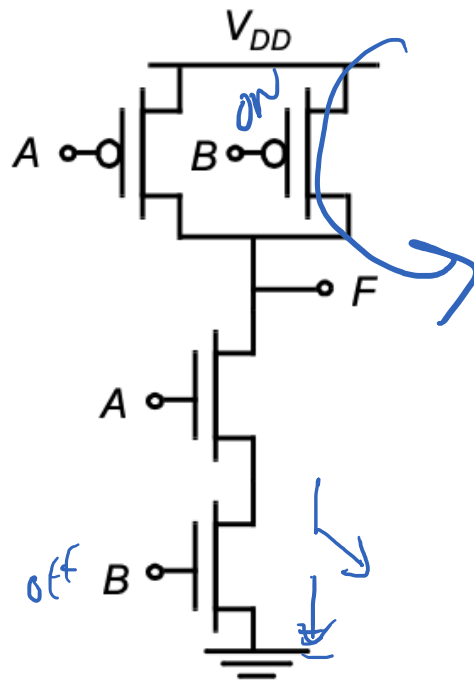


... and back to timing ...

NAND/NOR/INV



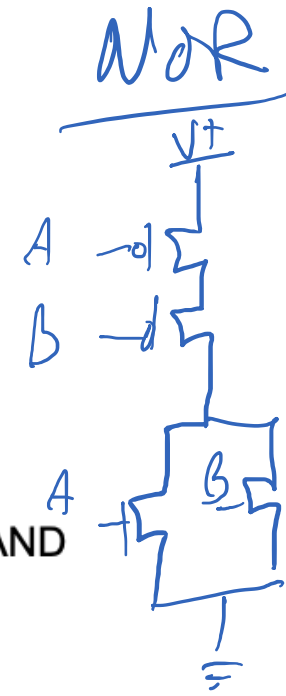
# NAND



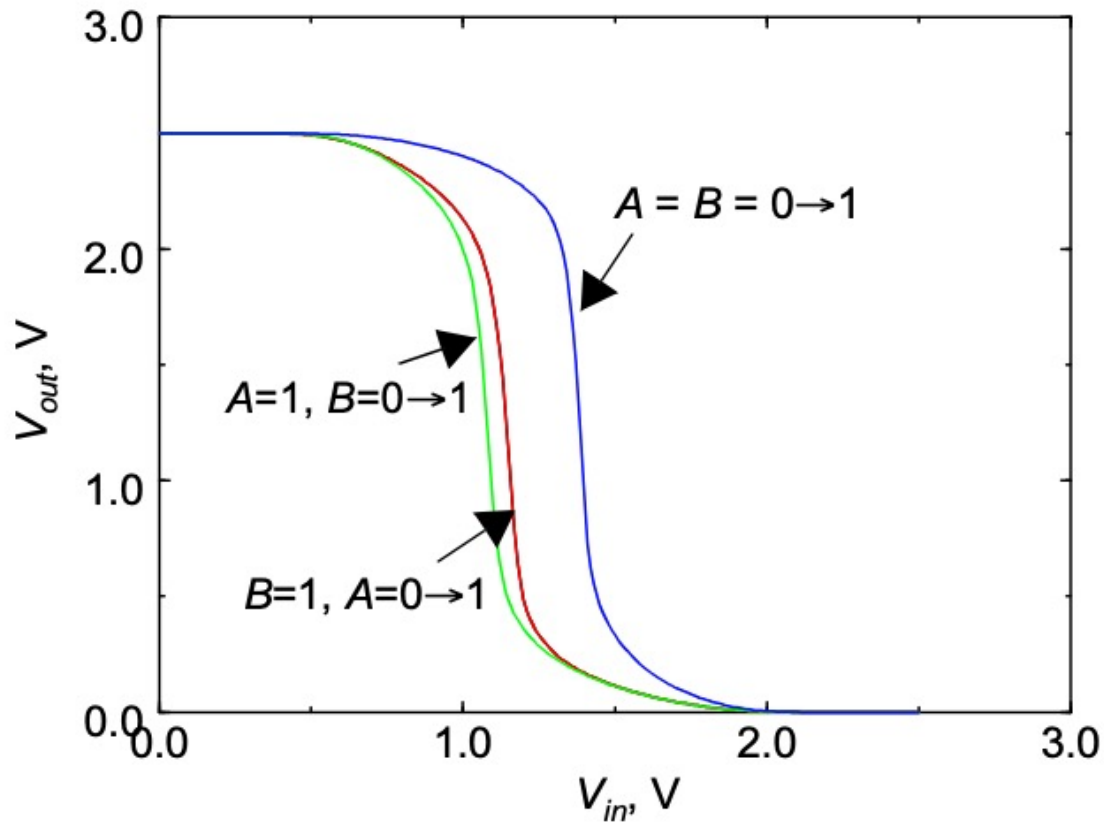
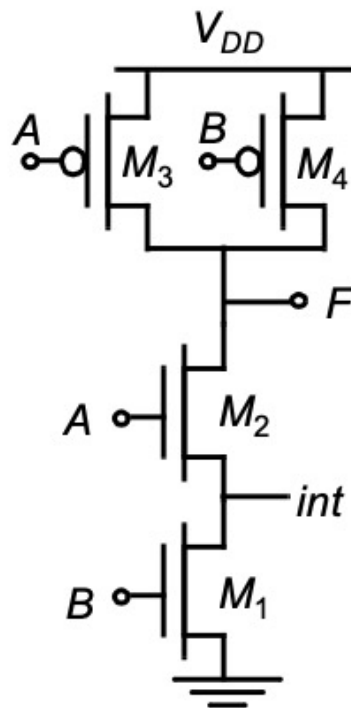
**Figure 6.5** Two-input NAND gate in complementary static CMOS style.

**Table 6.1** Truth Table for 2 input NAND

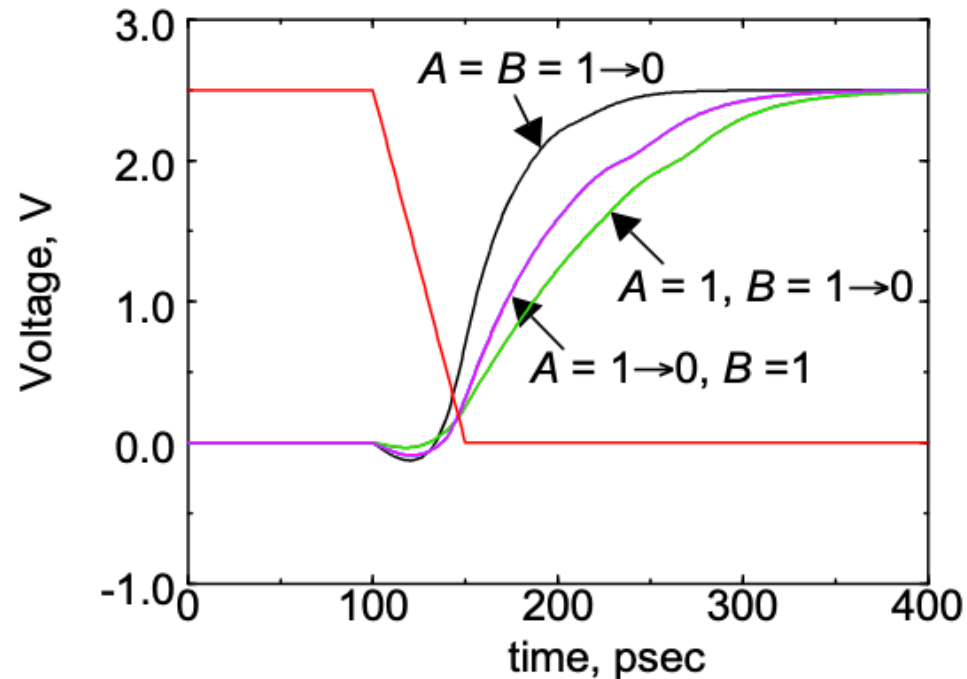
| A | B | F        |
|---|---|----------|
| 0 | 0 | <u>1</u> |
| 0 | 1 | <u>1</u> |
| 1 | 0 | <u>1</u> |
| 1 | 1 | 0        |



# Output delays are input dependent



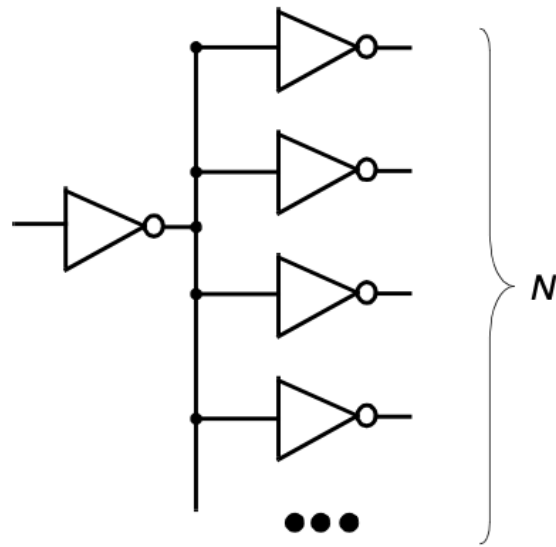
# Output delays are inputs dependent



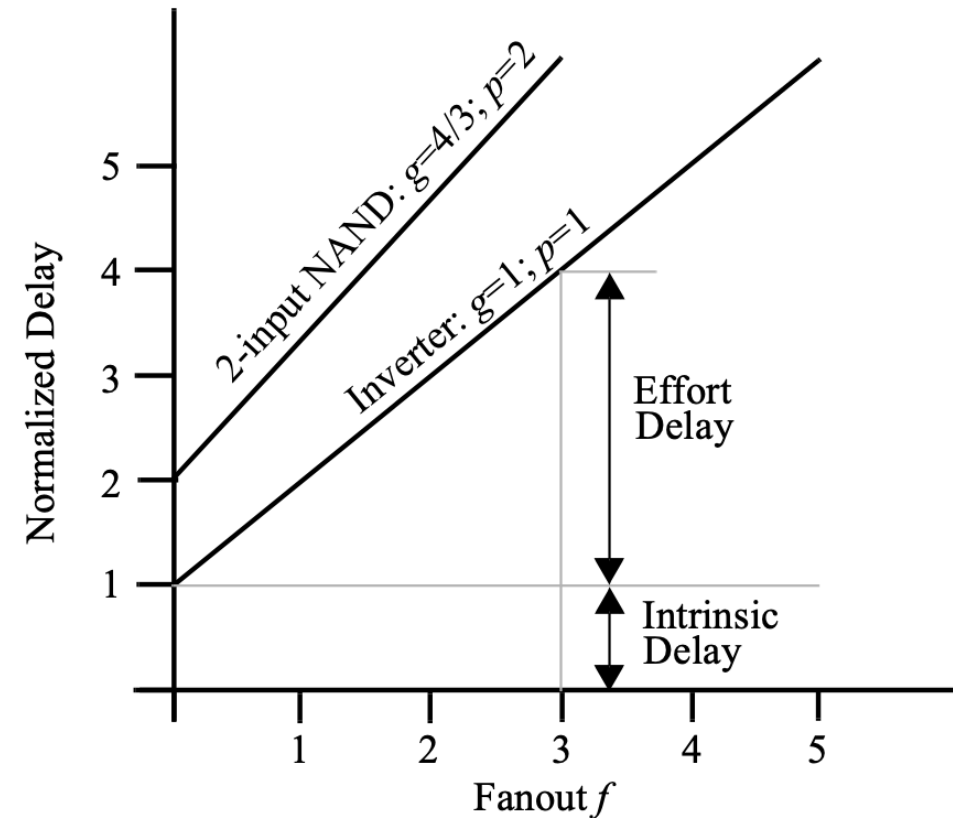
| Input Data Pattern           | Delay (psec) |
|------------------------------|--------------|
| $A = B = 0 \rightarrow 1$    | 69           |
| $A = 1, B = 0 \rightarrow 1$ | 62           |
| $A = 0 \rightarrow 1, B = 1$ | 50           |
| $A = B = 1 \rightarrow 0$    | 35           |
| $A = 1, B = 1 \rightarrow 0$ | 76           |
| $A = 1 \rightarrow 0, B = 1$ | 57           |

**Figure 6.9** Example showing the delay dependence on input patterns.

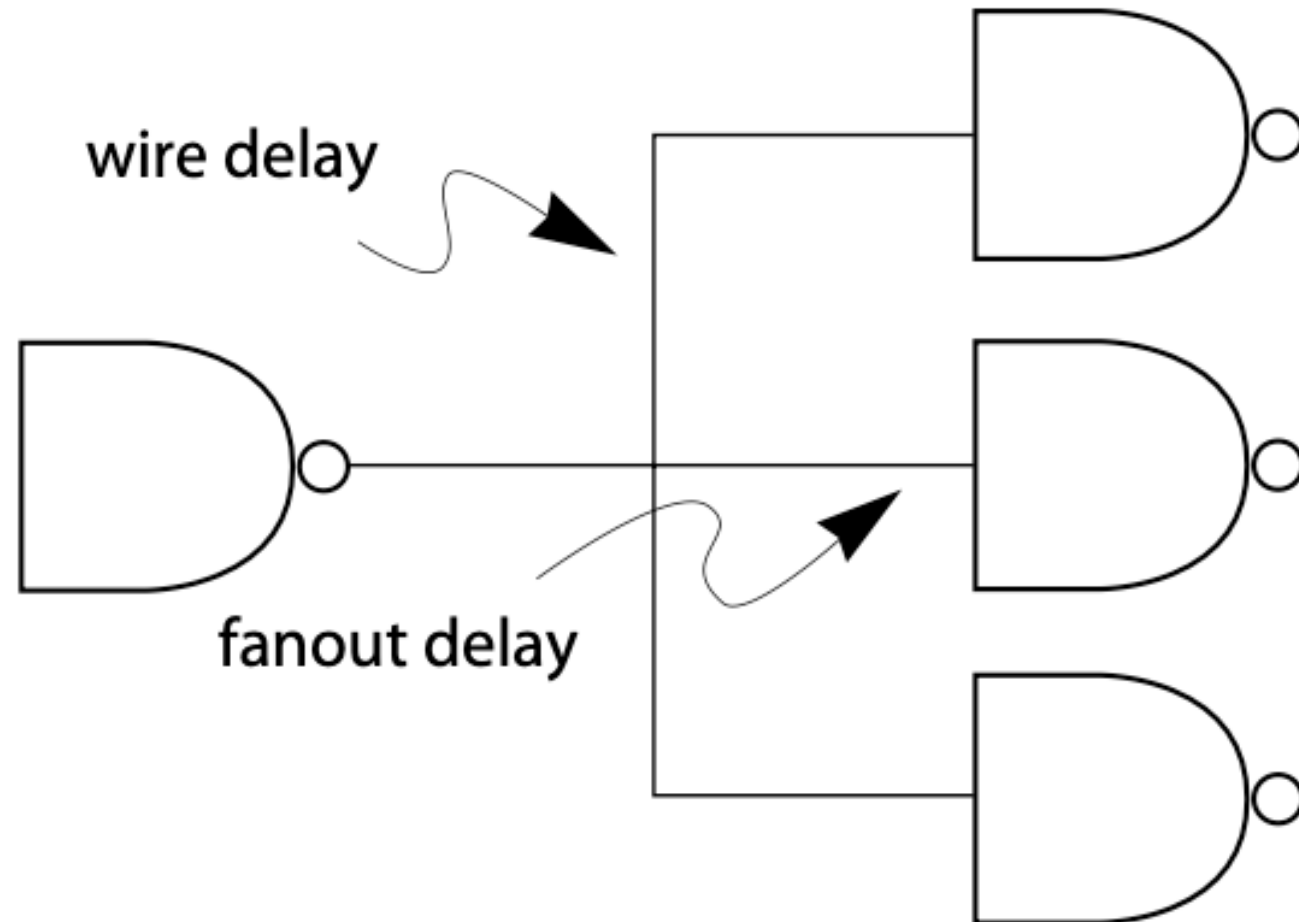
Output delays are also dependent on **Fan-Out**, or the number of dependent gates connected to driver output (capacitive load)



(a) Fan-out  $N$



Output delays are also-also dependent on wire delay (resistive + capacitive load).



# A Timing Library (.lib) defines the timing delays for each cell

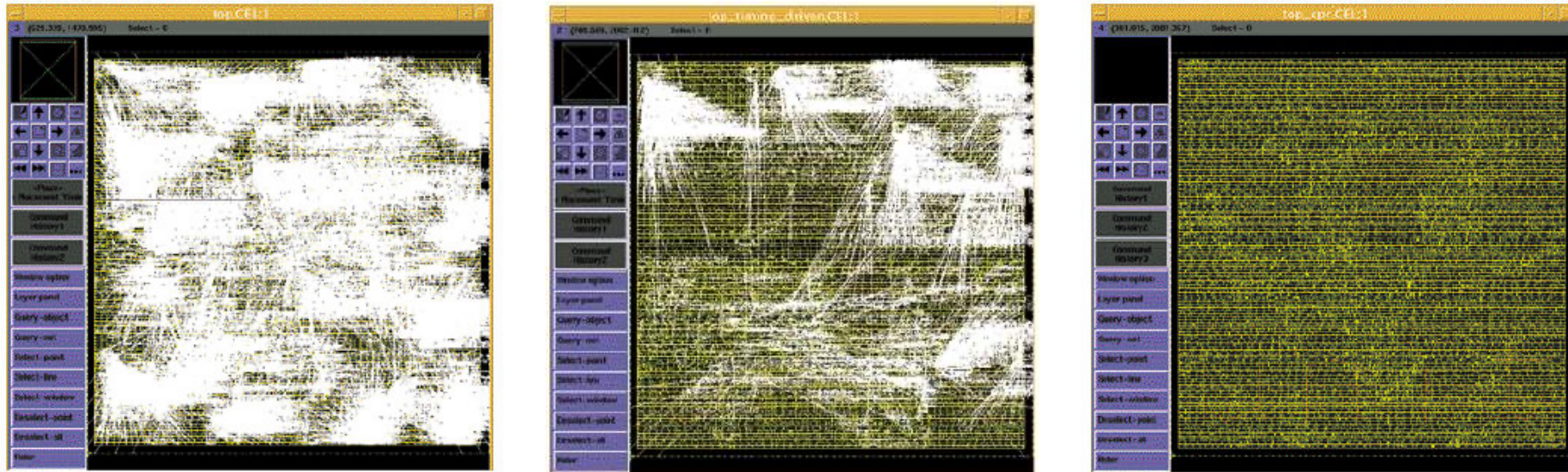
- Intrinsic Cell Delay
- Function of inputs
- Function of output load
  - Fanout load and wire load are lumped together

# Why do you care about per-cell timing?

- Critical path analysis
- Setup/Hold timing
  - Aka. Design Closure
  - Aka. Timing Closure



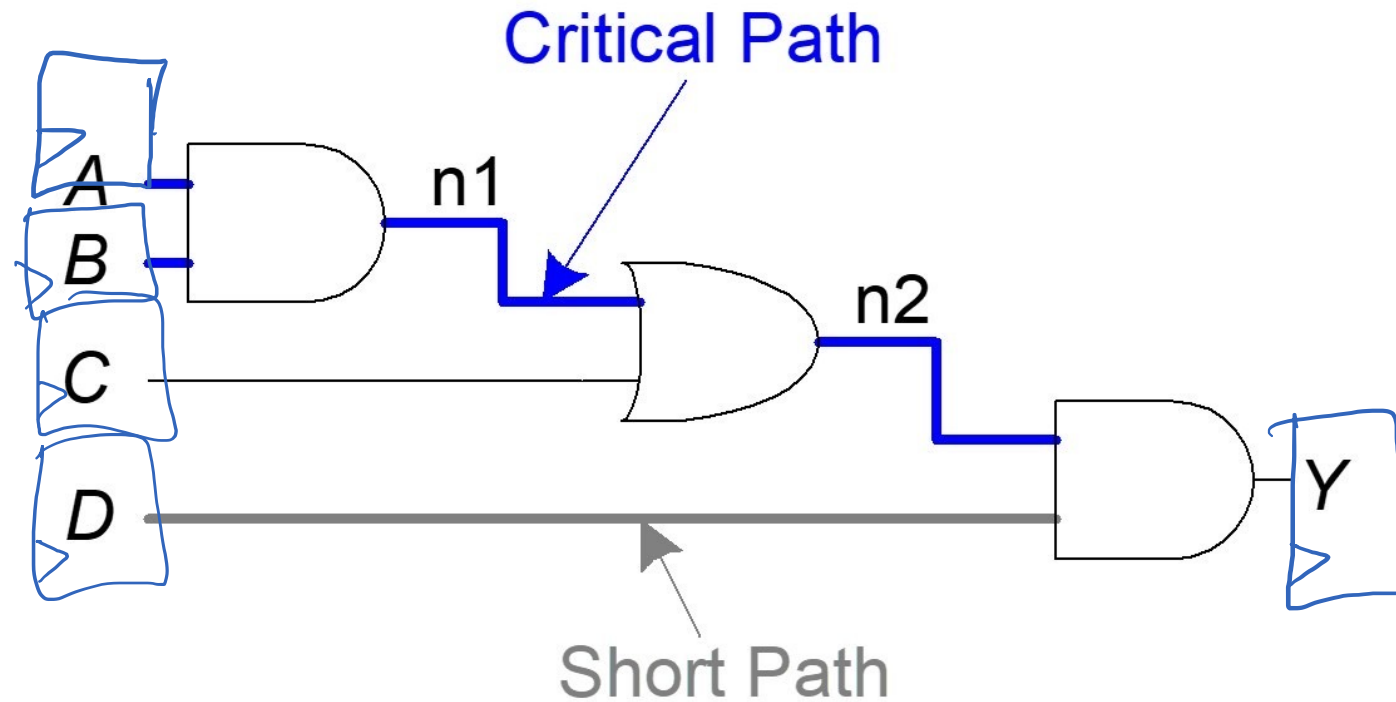
# The “Design Closure” Problem



Iterative Removal of Timing Violations (white lines)

# Critical Path

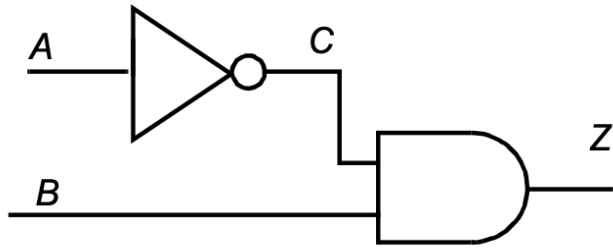
*longest delay path*



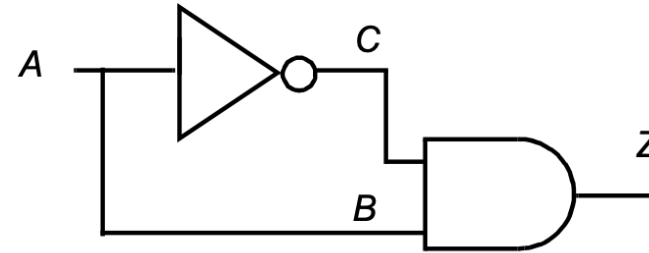
<https://www.linkedin.com/pulse/timing-combinational-circuits-neeraj-cheryala/>

# Reconvergence Paths

skip

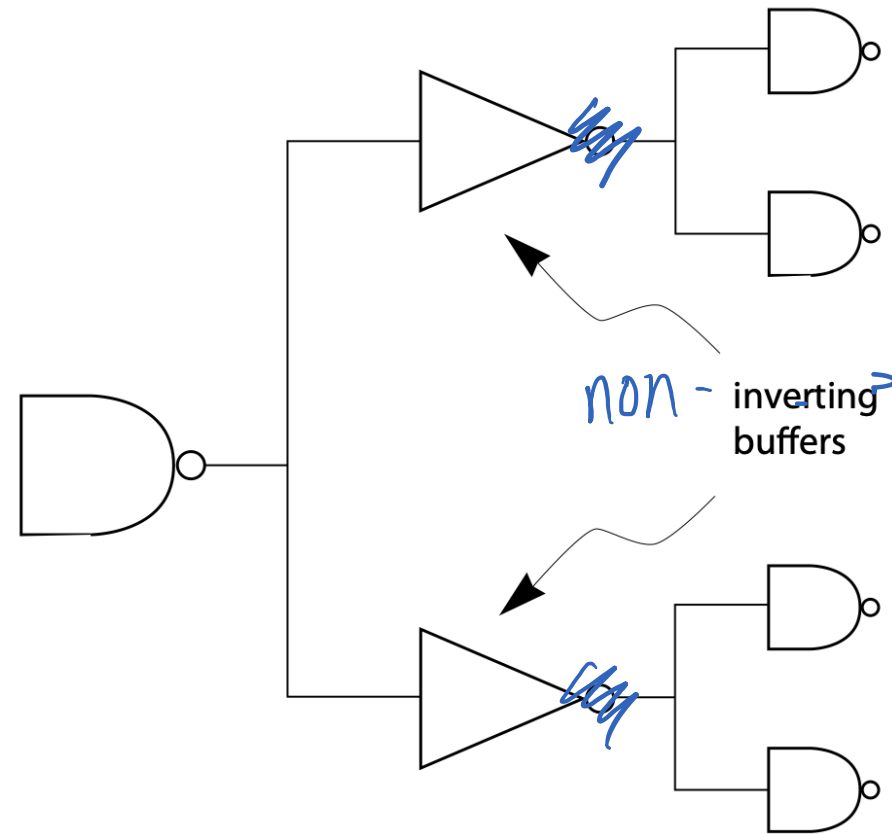


(a) Logic circuit without  
reconvergent fanout

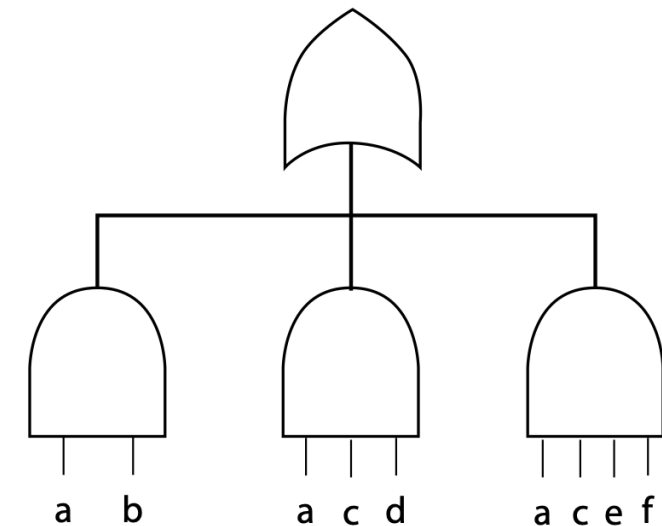
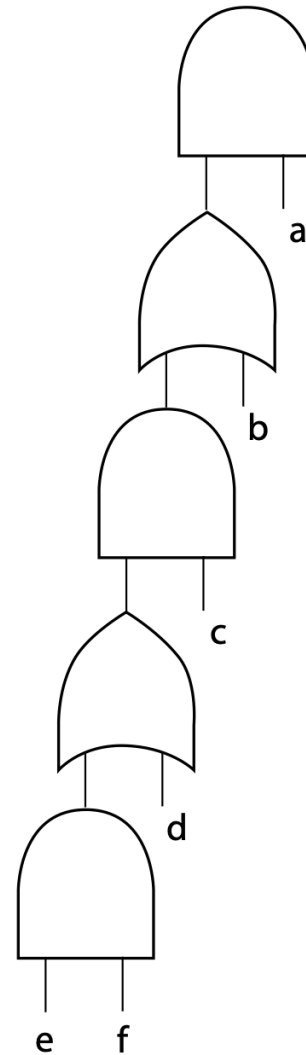


(b) Logic circuit with  
reconvergent fanout

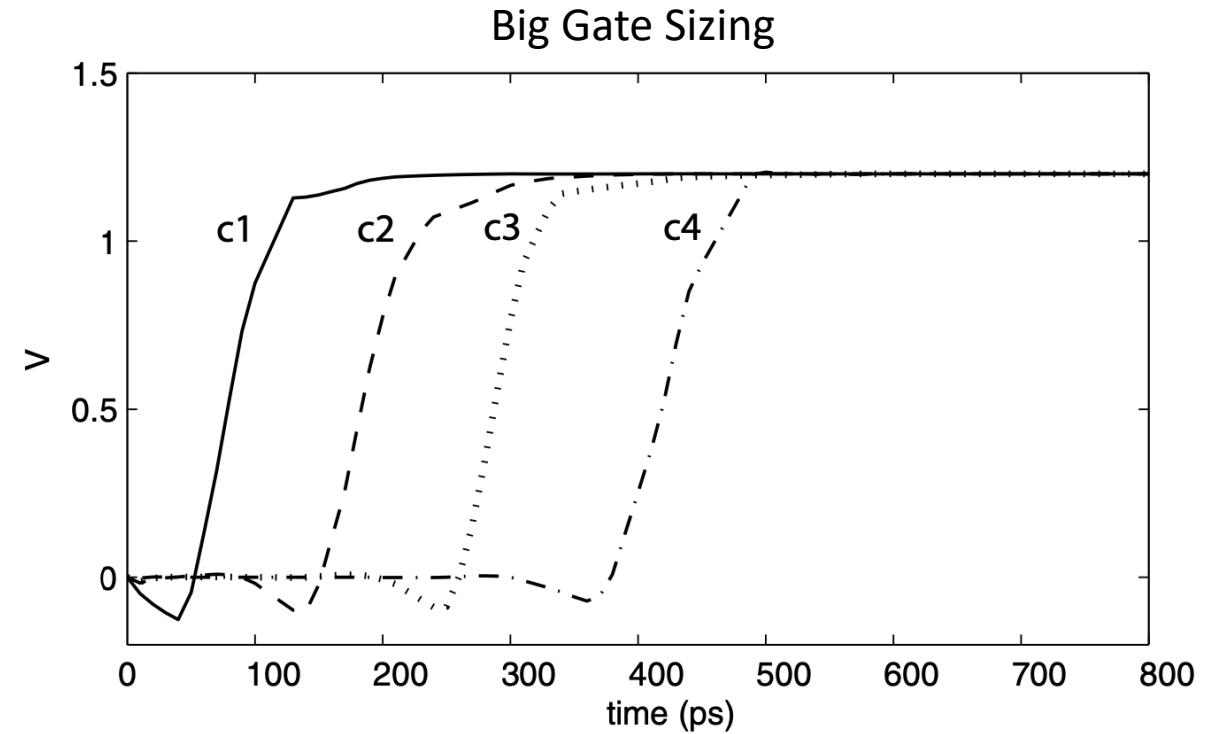
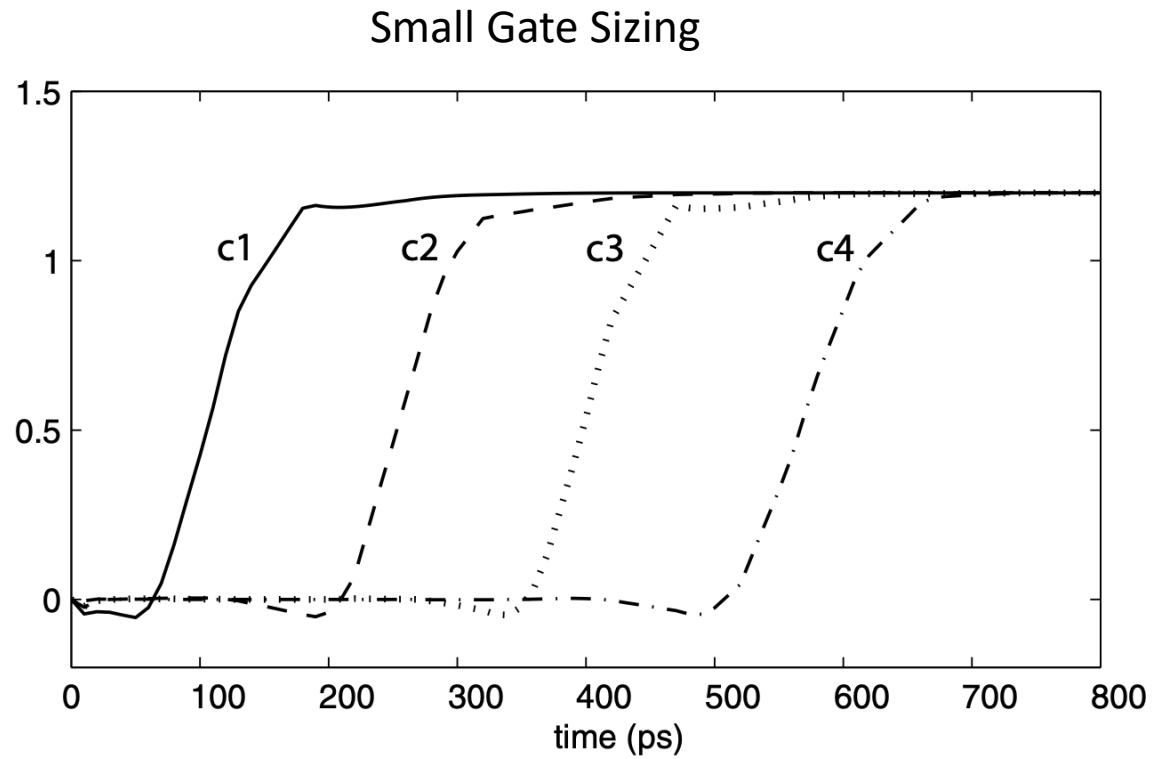
# Fan-out reduction can improve critical path timing



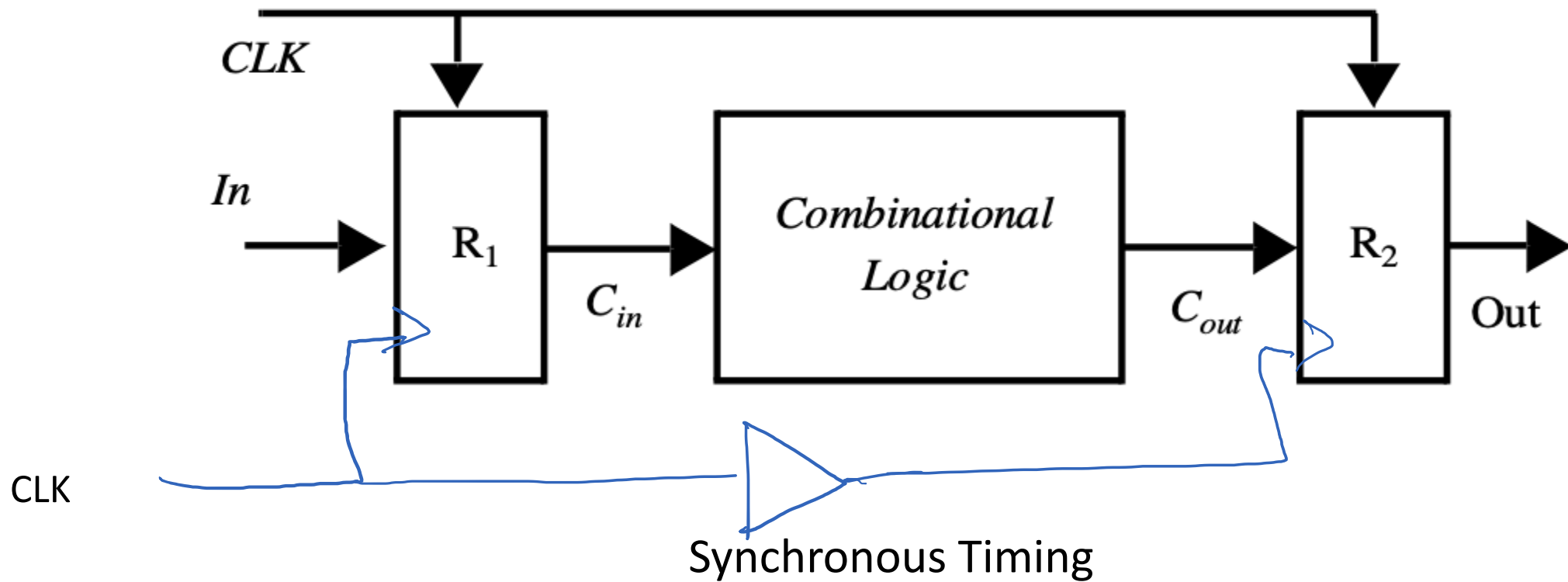
# Logic Simplification also improves critical path timing



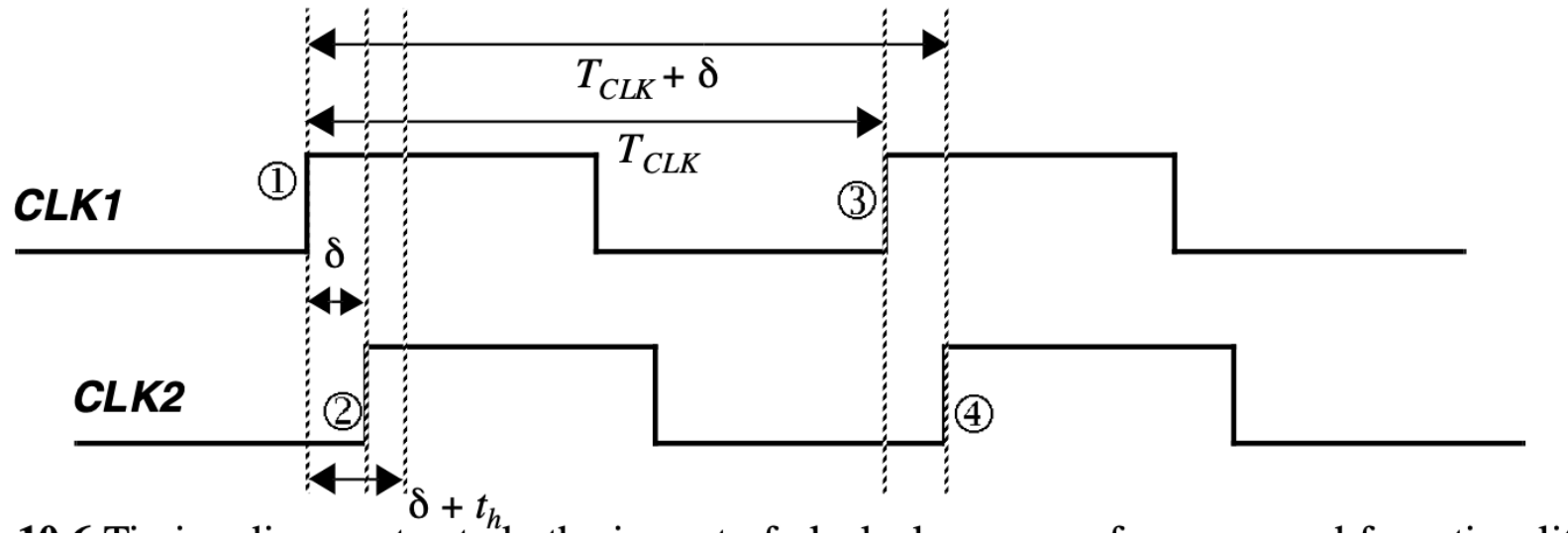
# Logic Cell sizing also improves critical path timing. To a point.



# Critical Path Timing with Flip-Flops (Registers)

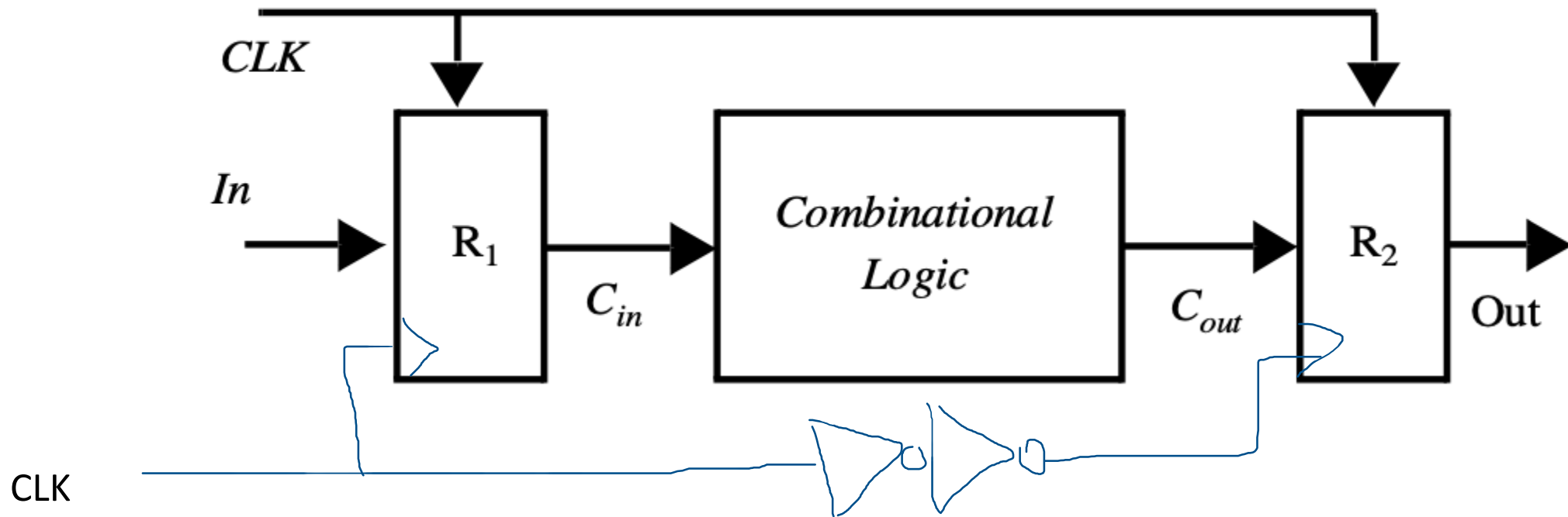


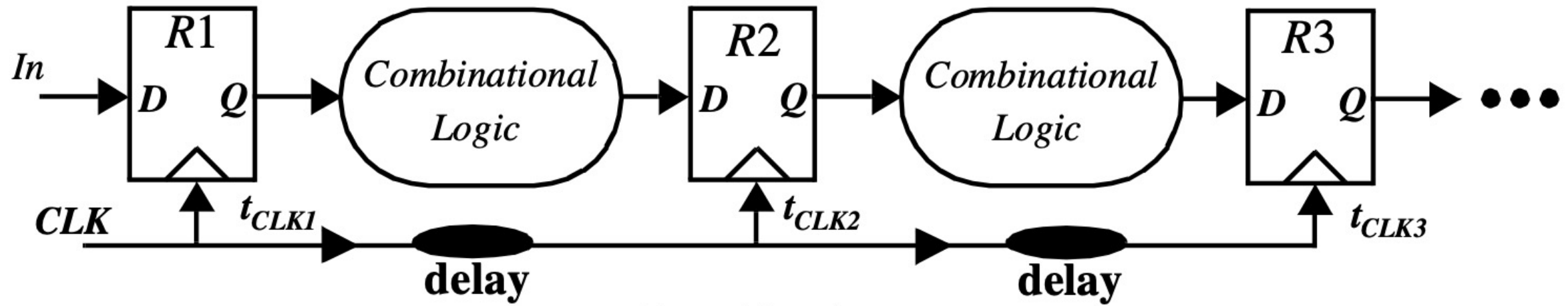
# Clock Skew does NOT improve timing...



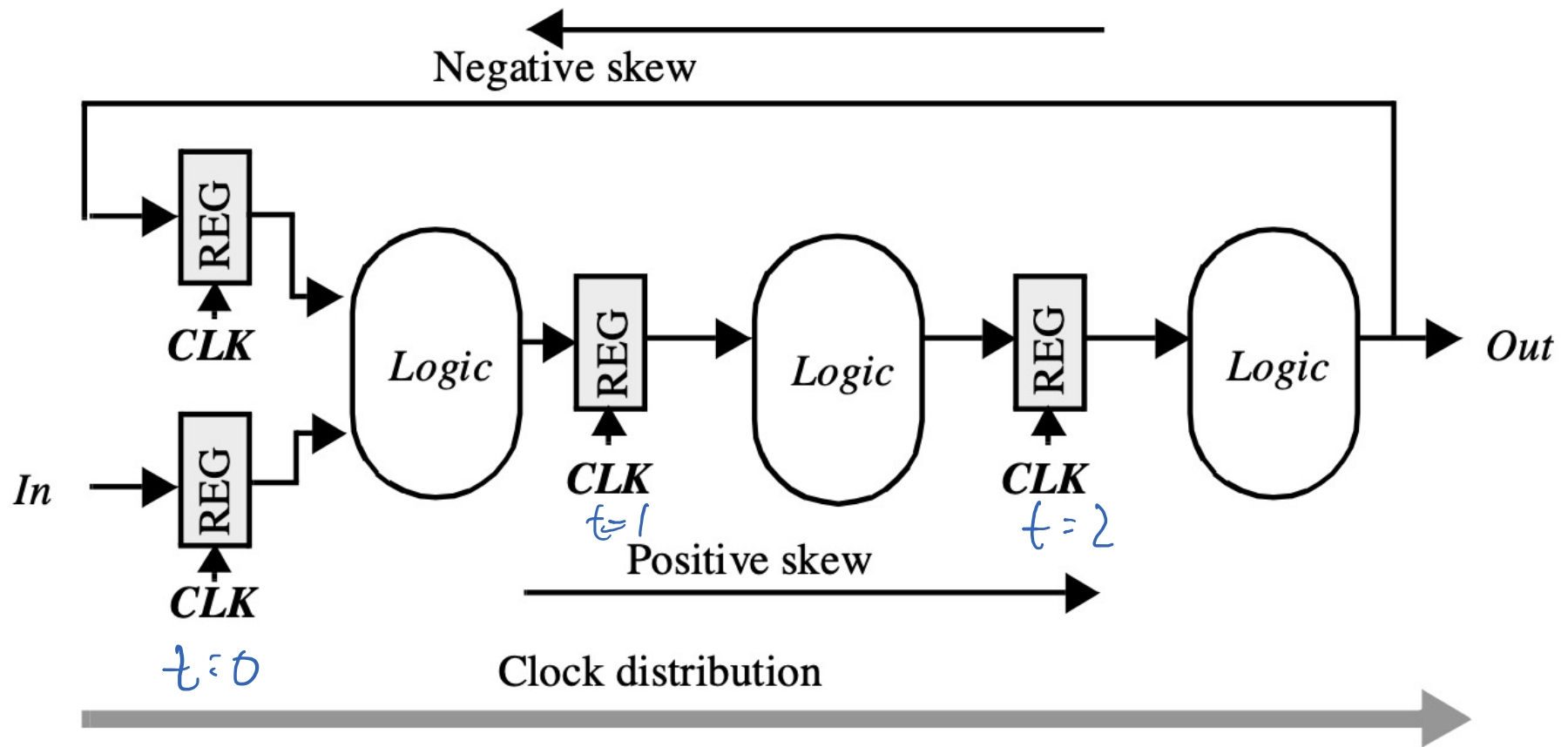


# Critical Path Timing with Clock Skew





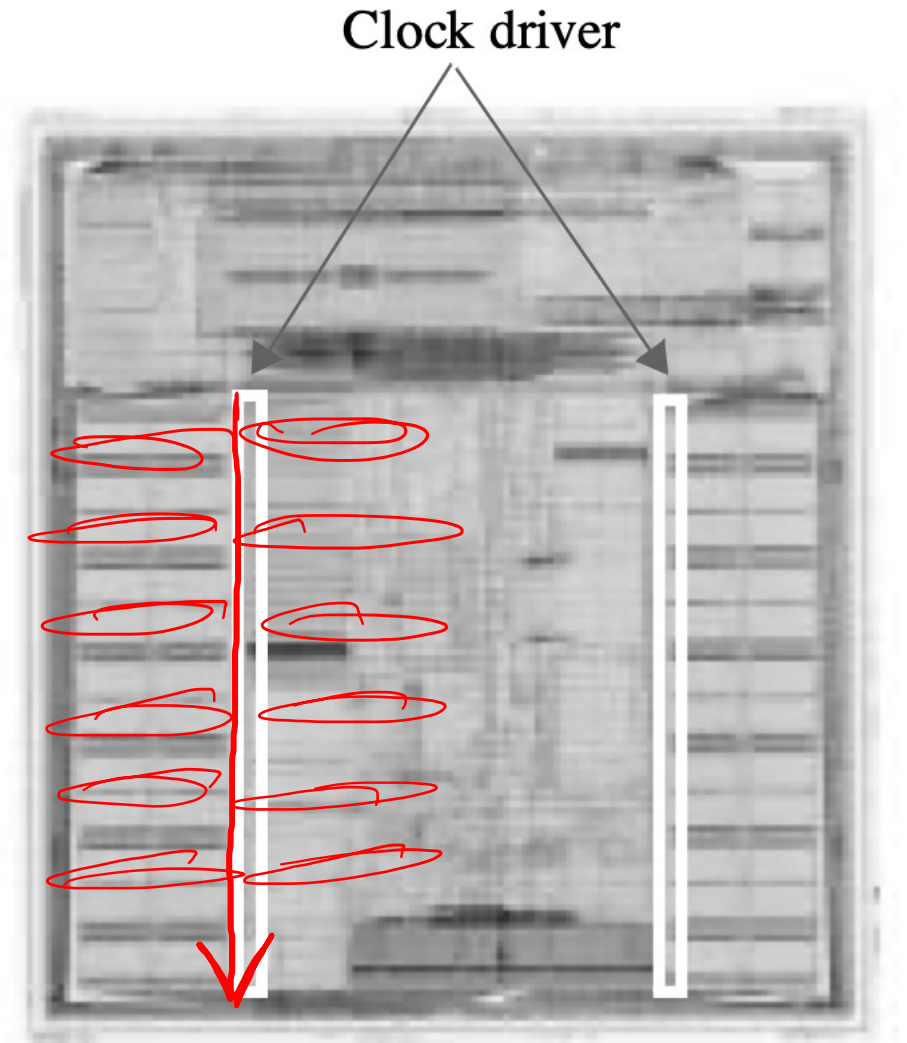
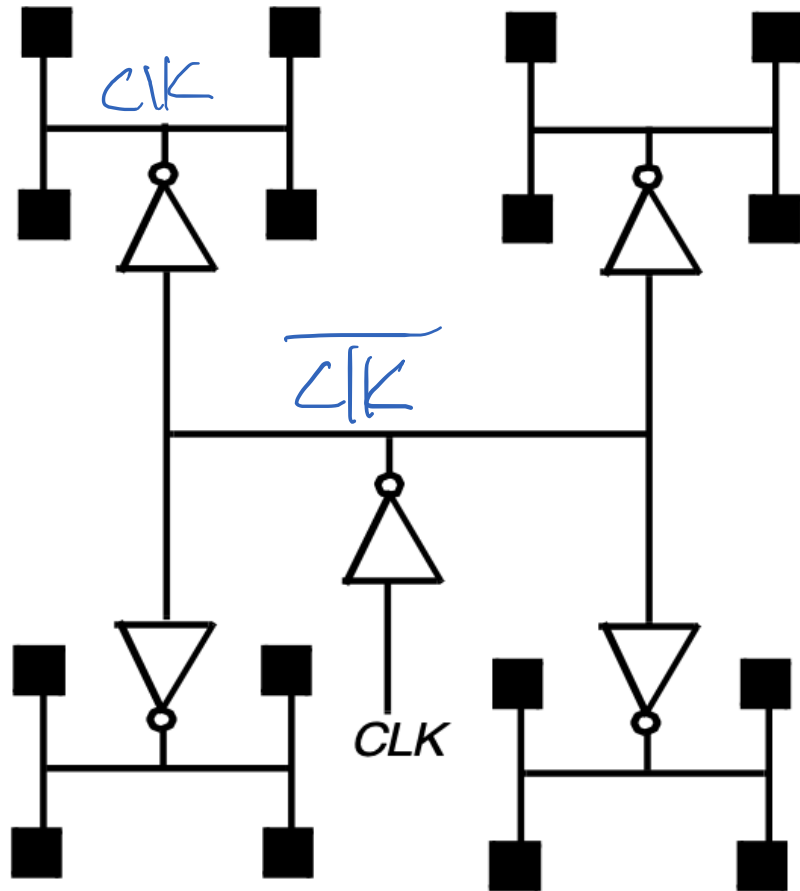
(a) Positive skew



# Clock Distribution to minimize skew

- H-Tree
- Fishbone

# H-Tree vs. Fish-Bone

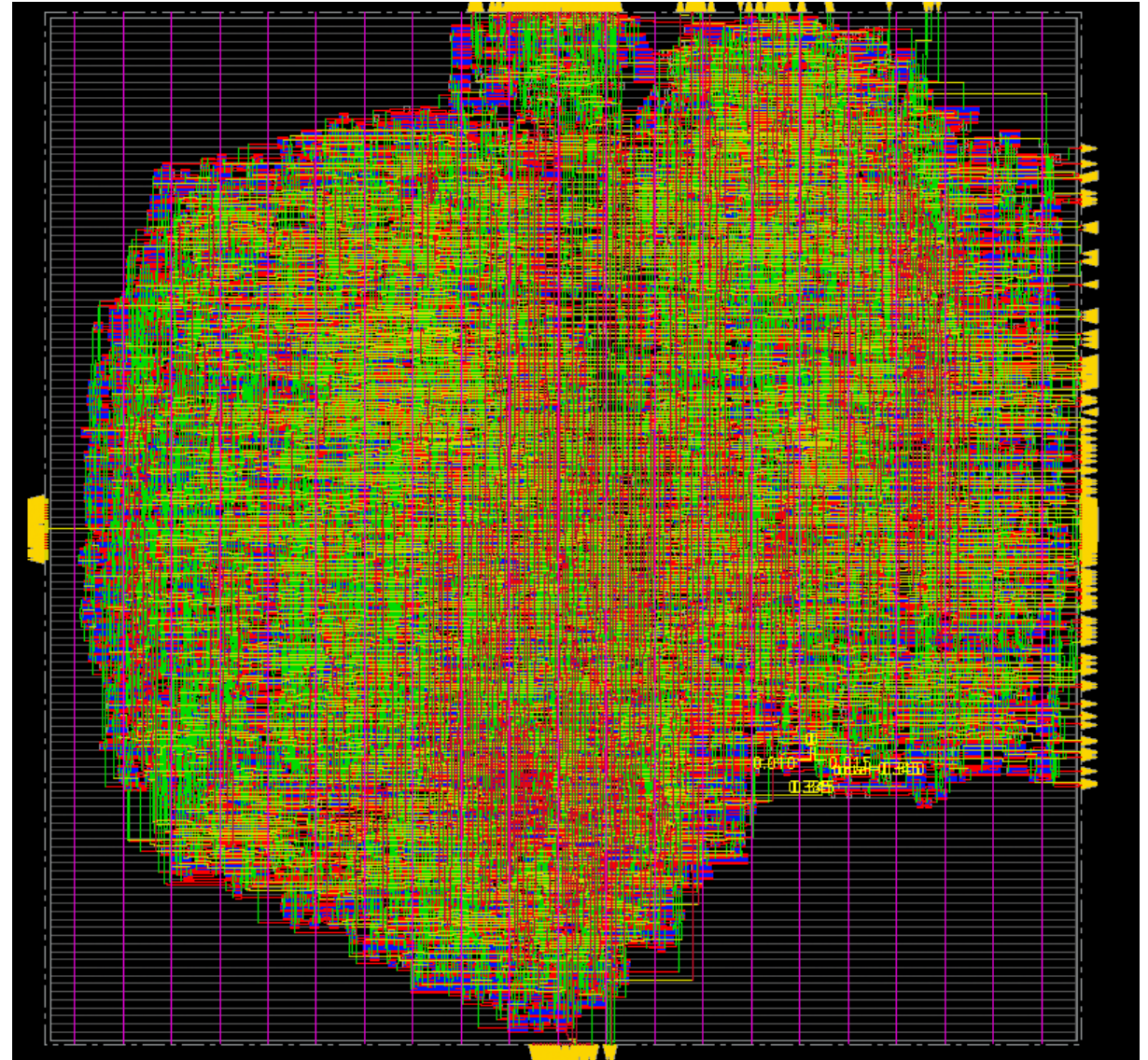


# List of interesting stuff

- PDK
- StdCell Library
- IO Cell Library
- Other Hard “Macros”
  - SRAM

# Project 1

- Timing Report



# Project 1 – make synth

- What is synthesis?

a process where an abstract description of a digital circuit (often at the register transfer level or RTL) is automatically translated into a gate-level implementation, optimized for specific design constraints

a.k.a “compiling”



# Project 1 – make pnr

- What is pnr?
- Place and Route
- Aka P&R, Automatic Place and Route (APR)

is where electronic components and their interconnections are automatically arranged and routed on a chip or printed circuit board (PCB)

# Report Timing

- Setup: `report_timing -early`
- Hold: `report_timing -late`
- `picorise`

*- add help docs  
- add 250mhz target  
- unify docs on reports*

# Things to try

## SkyWater Foundry Provided Standard Cell Libraries

There are seven standard cell libraries provided directly by the SkyWater Technology foundry available for use on SKY130 designs, which differ in intended applications and come in three separate cell heights.

Libraries `sky130_fd_sc_hs` (high speed), `sky130_fd_sc_ms` (medium speed), `sky130_fd_sc_ls` (low speed), and `sky130_fd_sc_lp` (low power) are compatible in size, with a 0.48 x 3.33um site, equivalent to about 11 met1 tracks.

Libraries `sky130_fd_sc_hd` (high density) and `sky130_fd_sc_hdll` (high density, low leakage) contain standard cells that are smaller, utilizing a 0.46 x 2.72um site, equivalent to 9 met1 tracks.

The `sky130_fd_sc_hvl` (high voltage) library contains 5V devices and utilizes a 0.48 x 4.07um site, or 14 met1 tracks.

# Things to try

## **TWO\_CYCLE\_COMPARE (default = 0)**

This relaxes the longest data path a bit by adding an additional FF stage at the cost of adding an additional clock cycle delay to the conditional branch instructions.

*Note: Enabling this parameter will be most effective when retiming (aka "register balancing") is enabled in the synthesis flow.*

## **TWO\_CYCLE\_ALU (default = 0)**

This adds an additional FF stage in the ALU data path, improving timing at the cost of an additional clock cycle for all instructions that use the ALU.

*Note: Enabling this parameter will be most effective when retiming (aka "register balancing") is enabled in the synthesis flow.*